# VERA System Documentation

Inner Workings, Architecture & Design Decisions

*This document explains HOW VERA thinks — the mechanics behind every intelligence feature, data flows between systems, and the reasoning behind key design decisions. For setup and configuration, see README.md. For how to use the dashboards, see the user guide, the cheat sheet and Victoria's lite guide in this folder.*

## Table of Contents


**1.** Design Philosophy
**2.** The Nightly Intelligence Pipeline
**3.** Flag System Deep Dive
**4.** Signal Learning & Noise Filtering
**5.** Cross-Domain Pattern Recognition
**6.** Pacing & Capacity Mode
**7.** Anticipator — Reminder Engine
**8.** Chat System Architecture
**9.** Slack Integration Architecture
**10.** Travel Pipeline
**11.** PTO & Clear Windows
**12.** Summaries Architecture
**13.** Monthly Life Review
**14.** Configuration Architecture
**15.** Trip Identity, Emails & Readiness
**16.** Packing, Luggage & Leaving Home
**17.** Finances (November 2026)
**18.** Places, Gifts, Life & Career
**19.** Family Planning & the Cycle Tracker
**20.** Dashboards, Navigation & Notifications
**21.** Your Assistants and How They Connect
**22.** Appendix: Developer Reference


## 1. Design Philosophy

### Why Google Apps Script?

Google Apps Script runs in Google's cloud with zero infrastructure cost and no deployment complexity. The alternative — a hosted server (VPS, Lambda, etc.) — would require managing uptime, auth, cron scheduling, and API proxying. Apps

Script eliminates all of that:

- **Native Google service access**: `CalendarApp`, `GmailApp`, `DriveApp`, `SpreadsheetApp` are first-class citizens with pre-authorised OAuth. No token management needed.
- **Built-in time-based triggers**: `ScriptApp.newTrigger('nightlyRun').atHour(23).everyDays(1)` replaces an entire cron + server setup.
- **No cold starts for the nightly batch**: The 11pm trigger fires reliably against Google's own infrastructure.
- **Execution logs**: Every `Logger.log()` call is persisted and viewable in the Apps Script execution log — no log aggregation service required.

The trade-offs are real: 6-minute execution time limit, no persistent process, no in-memory caching across executions. These are worked around with Script Properties (persistent key-value store) and per-execution caching via `_configCache_`.

## Why Google Sheets as the database?

Google Sheets serves as VERA's operational database. This is a deliberate choice, not a workaround:

- **Visible to the user**: Ahmed can open the sheet, see every flag, task, and metric in a grid, and manually edit any row. There is no hidden state.
- **No schema migrations**: Adding a new column to `TASK_HEADERS` requires updating the constant in `Code.js` and ensuring `ensureSheet()` writes the header on the next run. No SQL ALTER TABLE.
- **Tab-as-namespace**: Each domain (Flags, Tasks, Goals, Itinerary, GymLog, etc.) lives in its own tab. The `TABS` object in `Code.js` defines 70+ tab names — each is effectively a table.
- **Formula-driven dashboards**: The React dashboard reads data via the Web App API, but power users can apply Google Sheets formulas directly to any tab for custom analysis.
- **Audit trail by default**: Sheets preserves edit history natively. Every nightly write is visible in the revision history.

The schema for every tab is defined as a constant array in `Code.js` (e.g. `FLAG_HEADERS`, `TASK_HEADERS`, `GOAL_HEADERS`). The `ensureSheet()` function creates tabs with headers on first run and skips if they already exist — making the system safe to re-run `setupVERA()` without clobbering data.

## Why Claude AI for flags (not pure rules)?

Rule-based systems excel at precise conditions: "bill due in 5 days" or "task 14+ days old". But life is ambiguous:

- A meeting labeled "Team Sync" might require Ahmed to prepare a deck — or not. Context determines which.
- A task called "follow up with doctor" means something different if there is an upcoming health appointment on the calendar.
- Spotting that a bucket-list destination is adjacent to an upcoming trip requires geographic reasoning, not a simple date comparison.

Claude reasons over the full assembled context (calendar events, open tasks, summaries, PTO status, interest ledger, bills, home items, goals, trips, financial goals) in a single pass and surfaces what matters. The prompt includes explicit rules for each domain (see Section 2 / `buildPrompt()`) but Claude applies them with judgment, not literal pattern matching.

## Rule-based vs AI-assisted split

The split is intentional and deliberate:

| Approach | Used for |
|---|---|
| Rule-based | Escalation aging (3d/7d thresholds), pacing mode miss scoring, pattern recognition triggers, reminder cooldowns, flight status polling, PTO calculation, bill due-date detection |
| AI-assisted (Claude) | Nightly flag generation, chat responses, packing list generation, weekend planning memos, monthly review synthesis, Explorer daily discovery |

Rule-based checks run first, every night, regardless of AI availability. If the Claude API call fails, all the rule-based flags and Slack logs still go out. The Claude call is the only step that can produce zero output on a network error — everything else is local computation.

### Why no build pipeline?

The React dashboard (`index.html`) uses Babel Standalone loaded from a CDN. This means:

- No `npm install`, no `webpack`, no build step.
- The file is a single HTML file that can be opened locally, pushed to GitHub Pages, or deployed to Netlify as a static site.
- The entire frontend state is managed with React hooks; no Redux, no router.
- Dark VERA theme is pure CSS variables defined in a `<style>` block.

The trade-off is a slightly longer initial parse time for Babel in the browser. For a personal dashboard used by two people, this is acceptable.

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## 2. The Nightly Intelligence Pipeline

`nightlyRun()` fires at 11pm daily via the Apps Script time-based trigger installed by `setupTriggers()`. It is the orchestration function: it runs every pre-computation step in order, then calls Claude, then writes flags.

All pre-computation steps (Steps -1 through 0q) are wrapped in individual `try/catch` blocks. Failures are pushed into a `stepFailures` array and reported to `#vera-logs` at the end of the run — they do not abort the pipeline. Steps 1–4 (data collection, Claude call, flag write) are inside a single outer `try/catch`; a failure here sends an error email to `CONFIG.MORNING_NUDGE_EMAIL` and a failure message to `#vera-logs`.

### Step -1: `escalateAgedFlags_()`

Scans every unresolved, unacknowledged flag in the `Flags` tab. For each flag:

- If age >= 3 days and `Escalated` column is empty: bumps urgency up one level (Low→Medium, Medium→High, High stays High) and writes `'3d'` to the `Escalated` column (col K).

- If age >= 7 days and `Escalated` column is `'3d'`: appends a stale note to the `Reason` field and writes `'7d'` to `Escalated`.

Snoozed flags are skipped while `Snoozed Until` is a future date. Each threshold fires exactly once per flag — the `Escalated` column tracks state.

### Step 0: `writeSummarySnapshot()`

Calls `writeMetrics_(ss)` and `writeSummaries_(ss)` in sequence. Both operations clear `[AUTO]`-prefixed rows from their respective tabs and rewrite them fresh. This runs before Claude is called so the AI sees current data.

- `writeMetrics_()` writes task counts, calendar density, and flag counts to the `Metrics` tab.
- `writeSummaries_()` reads `summary_sheet:SourceName` Config rows and pulls live cell values from external Google Sheets.

This step is non-fatal at the outer level but each individual tab write is wrapped internally.

### Step 0a: `syncCalendarBirthdaysToImportantDates_()`

Reads upcoming birthday events from the Joint Chaos calendar (events matching a birthday keyword pattern) and writes them to the `Important Dates` tab if not already present. This keeps the People tab's Important Dates populated automatically without manual entry.

### Step 0a-ii: `resetChoresByCadence_()`

Scans the `Chores` tab. For each chore, checks whether enough time has elapsed since `Checked At` based on the `Cadence` column (Daily/Weekly/Bi-weekly/Monthly/Quarterly). If the cadence interval has passed, resets `Checked` to false so the chore appears unchecked in the dashboard for the new period.

### Step 0b: `writePTOSnapshot_()` → returns `ptoStats`

Reads Ahmed's work calendar (default: `Verizon Calendar`) and classifies all-day events as vacation days, personal time, or company holidays based on title keywords configured in the Config tab. Computes days used, days remaining, burn-rate vs ideal pace, and upcoming travel windows. Writes `[AUTO]` rows to the `Summaries` tab. Returns a `ptoStats` object that is passed downstream to the Monthly Review (Step 0o) and the Claude prompt to avoid re-reading the PTO data twice.

### Step 0c: `runExplorer_()`

Calls Claude in "discovery mode" — a separate prompt (built by `buildExplorerPrompt_()`) that asks Claude to surface 2–3 non-obvious, forward-looking observations connecting goals, tasks, calendar, and the Shared Interest Ledger. Uses a date-keyed entry in `Reminders Memory` to fire at most once per day. Output is delivered via `sendNudge_()` — Slack if configured, email otherwise.

### Step 0d: `getSuppressedKeyPatterns_()`

Reads the `SignalLearning` tab and returns all patterns where `Suppressed = Yes`. The returned array is passed into `generateFlags()` (Step 2&3) and injected into the Claude prompt as a "SUPPRESSED TOPICS" list — patterns the user has consistently snoozed or ignored will not be re-flagged. See Section 4 for the full scoring logic.

### Step 0e: `recordExpiredFlags_()`

Scans the Flags tab for flags that are open (unresolved, unacknowledged, unsnoozed) and either: (a) older than 30 days, or (b) marked `Escalated = '7d'`. For each, calls `recordFlagOutcome_(flagKey, 'expired')` to add a noise-score penalty in Signal Learning. Uses the `Escalated = '7d'` marker as a proxy for "VERA tried, user never responded."

### Step 0f: `checkPreTripBriefings_()`

Reads the `Itinerary` tab, groups rows by trip key (`YYYY-MM-DD|Trip Label` format), and finds trips whose departure date is within the configured `pretrip_briefing_hours` window (default: 48 hours). For each qualifying trip, assembles a structured briefing flag (via `buildPreTripBriefingFlag_()`) covering weather, packing completion, flight status, itinerary items, and confirmation numbers. Deduplication is handled by `writeFlags()` — the flag key ensures one briefing per trip, per lifetime.

### Step 0g: `checkPostTripCapture_()`

Fires `posttrip_capture_delay_days` (default: 1) days after a trip's end date. Writes a flag prompting Ahmed to log trip memories, expenses, and notes. Gated by `posttrip_capture_enabled` Config key.

### Step 0h: Morning routine reset

Directly reads the `Morning Routine` tab and sets `Checked = false` and `Checked At = ''` for every item row (columns 5–6). This resets the daily checklist so it starts fresh each morning. Intentionally a direct sheet write, not a function call — it is simple enough to not warrant its own module.

### Step 0i: `checkGymSessions_()`

Reads the `Gym Log` tab and the calendar within the configured `gym_tracker_lookback_hours` window (default: 24 hours). Looks for calendar events with titles matching gym/workout patterns that have ended. Writes new rows to `Gym Log` when found. Gated by `gym_tracker_enabled = 'true'` Config key.

### Step 0j: `checkFitnessConsistency_()` + `checkFitnessTravelGap_()`

`checkFitnessConsistency_()` evaluates gym session count vs the `fitness_weekly_target` Config key. If behind pace by the configured `fitness_low_flag_day` (default Wednesday = day 4), writes a Low-urgency flag. `checkFitnessTravelGap_()` checks if a trip is approaching and automatically creates gym session placeholder events in the VERA calendar for the travel period. Both are gated by `fitness_enabled = 'true'`.

### Step 0k: `autoRestockItems_()` + `generatePantryFlags_()`

`autoRestockItems_()` uses exponential moving average (EMA, configurable alpha via `pantry_ema_alpha`, default 0.3) to predict when household items will run out based on `Purchase History` tab data, and pre-adds restock items to the Shopping List. `generatePantryFlags_()` checks if a trip is approaching and items are predicted to run out during the trip. Both are gated by `pantry_enabled = 'true'`.

### Step 0l: `inferCapacityMode_()`

Reads tomorrow's calendar and scores load based on meeting count and event density. Writes a capacity mode signal (`busy`, `normal`, or `light`) to Script Properties. The `morningNudge()` function reads this to filter which flags to surface in the morning briefing — `busy` days show only High + time-sensitive flags.

### Step 0m: `checkContracts_()`

Reads the `Contracts` tab. For each active contract, checks whether the end date is within the notice period (stored in `Notice Period Days` column). Generates Medium flags for contracts approaching their notice window and High flags for expired contracts. Uses source `'General'` and key format `contract_renewal_[name_snake]`.

### Step 0n: `checkHealthAppointments_()`

Reads Google Calendar for events prefixed with `DR:` (the convention for health appointments). Computes days until the appointment and compares against configured intervals. Generates flags for overdue or upcoming appointments. This replaces the former `HEALTH_APPOINTMENTS` sheet tab — appointment data now lives in the calendar, not a sheet.

### Step 0o: `checkMonthlyReview_(ptoStats)`

Fires only on the 1st of each month. Checks whether a review for the prior month already exists in the Flags tab (dedup by key `monthly_review_YYYYMM`). If not, calls `buildMonthlyReview_()`, writes a Low-urgency flag, archives to the `Monthly Reviews` tab, and sends a styled HTML email. The `ptoStats` object computed in Step 0b is passed in to avoid re-reading PTO data. See Section 13 for the full assembly.

### Step 0q: `checkCrossPatternFlags_()`

Builds a cross-domain signal snapshot and evaluates 7 compound patterns. See Section 5 for the full pattern definitions and trigger conditions. Writes at most `pattern_max_flags` (default: 2) new flags per run.

### Step 1: Data collection

```
events    = getUpcomingEvents()       // Calendar.js — reads all Google Calendars
tasks     = getOpenTasks()            // Tasks.js — reads Tasks tab + Google Tasks
summaries = getSummaries()            // Summaries.js — merges Metrics + Summaries tabs
ledger    = getSharedInterestLedger_() // Interests.js — reads Shared Interests tab
```

`getUpcomingEvents()` reads the number of days configured in `calendar_days_ahead` (default: 7). It reads ALL calendars via `CalendarApp.getAllCalendars()`, applying `skip_calendars` Config rows to exclude noise calendars. Each event gets a `calLabel` from `calendar_label:CalName` Config rows.

### Step 1b: `suggestDueDates(tasks)`

For tasks with no due date, Claude suggests a due date based on task content and writes it back to the sheet. This is a separate, non-fatal Claude call made before the main flag generation call.

### Step 2: Early exit if no data

If `events.length === 0 && tasks.length === 0 && summaries.length === 0`, the Claude call is skipped entirely. A log entry and a Slack message are sent (`0 flags — no data`). This prevents burning API tokens on empty runs (e.g. quiet travel weekends with no tasks).

### Steps 2 & 3: `generateFlags()` — Claude API call

`generateFlags()` in `Claude.js` calls `buildPrompt()` to assemble the full context string, then calls the Anthropic API. The `suppressedPatterns` array from Step 0d is passed in and injected as a "SUPPRESSED TOPICS" block in the prompt. The response is expected to be a raw JSON array — `parseFlags()` strips any markdown fences and JSON-parses the result.

### Step 4: `writeFlags(flags)`

Persists Claude's output to the `Flags` tab. See Section 3 for the full dedup and fingerprinting logic.

### Step 4b: `recordFlagsGenerated_(flagKeys)`

Extracts the `key` field from each written flag and calls `recordFlagsGenerated_()` in `SignalLearning.js`, incrementing the `Total Seen` counter for each pattern. This feeds the noise-scoring loop described in Section 4.

### Slack summary: `sendSlackLog_()`

At the end of `nightlyRun()`, a summary is posted to `#vera-logs`:

```
■ Nightly run — 5 flags written (2 High, 2 Med, 1 Low) · 1 step warning in 47s
```

If `stepFailures` is non-empty, a second message lists each failing step. If the outer try/catch catches a fatal error, it sends a `■ Nightly run FAILED` message with the error and file/line.

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## 3. Flag System Deep Dive

### Flag creation — `writeFlags()`

`writeFlags(flags)` is called from `nightlyRun()`, `checkCrossPatternFlags_()`, `checkPreTripBriefings_()`, `checkMonthlyReview_()`, `checkMissRate_()`, `EmailParser.js`, and other rule-based checkers. It is the single write path for all flags.

Each flag object must contain: `source`, `flag`, `reason`, `urgency`, and optionally `key`.

**Fingerprinting and dedup**: Before writing, `writeFlags()` builds a `Set` of fingerprints from ALL existing flags (regardless of state — open, acknowledged, snoozed, or resolved). Once a key has been written in any state, it is permanently blocked from re-appearing. If the issue genuinely recurs, Claude is expected to generate a new key (e.g. adding a `_q2` suffix).

The fingerprint is computed by `makeFlagFingerprint_(source, flagText, key)`:

- If a stable `key` is present: fingerprint = `'key:' + safeKey` (key alone, no source prefix)
- If no key (legacy rows): fingerprint = `source + '|' + first 8 words of flag text`

**Fuzzy dedup** via `keysAreSimilar_()`: For key-bearing flags, checks whether the new key shares >= 60% of its meaningful tokens with any existing key fingerprint in the set. Month names and standalone numbers are stripped before comparison, so `verizon_bill_march_13` and `verizon_bill_march_14` are treated as the same issue. This prevents date-drifted duplicates when a recurring bill flag shifts by one day.

**Flag ID format**: `FLAG-YYYYMMDD-NN` where `NN` is a random two-digit number (10–99) chosen at write time. The randomness prevents collision when multiple flags are written in the same nightly run.

**Color coding**: After writing, `colorCodeFlags()` applies background colors to all rows in the Flags tab:

- High = `#fce8e6` (soft red)
- Medium = `#fef9e7` (soft yellow)
- Low = `#e6f4ea` (soft green)

## Flag key convention

Keys are stable, snake_case identifiers designed by Claude. The convention is `domain_description_context`, for example:

- `health_appt_ahmed_annual_physical`
- `cross_high_stress_compound`
- `verizon_bill_march_13`
- `task_neglect_TASK_0012`
- `fgoal_emergency_fund_at_risk`
- `packing_not_started_austin_trip`

Keys serve as the cross-run dedup anchor. When Claude generates the same conceptual flag on consecutive nights (e.g. a bill that is still unpaid), the key should match the existing flag in the sheet, causing `writeFlags()` to skip it as a duplicate.

## Escalation — `escalateAgedFlags_()`

Escalation runs at the very start of `nightlyRun()` (Step -1) so the urgency of aged flags is correct before the morning nudge reads them. The `Escalated` column (col K, index 10) stores `'3d'` and `'7d'` as watermarks so each threshold fires exactly once per flag lifetime. Snoozed flags are excluded from escalation while the snooze date is in the future. Nothing escalates while Away mode is on; a flag's age keeps counting, so anything still open escalates on the first night back. Column L, **Due**, holds a flag's deadline (`yyyy-MM-dd`) when its source knows one: bills (from Claude), contracts, coupons, card perks and travel documents. Away mode uses it.

### Flag lifecycle

```
created (Open)
    ↓
acknowledged (Acknowledged = 'Yes') OR
snoozed (Snoozed Until = future date) OR
resolved (Resolved = 'Yes')
```

Snoozed flags re-surface automatically: `writeFlags()` does not block them, and `escalateAgedFlags_()` skips them while snoozed — but the flag is still visible in the sheet and the dashboard. When the snooze date passes, the flag becomes active again.

When a user resolves a flag (via the dashboard or Slack button), `recordFlagOutcome_(flagKey, 'resolved')` is called, adding a positive signal score to the pattern in Signal Learning.

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## 4. Signal Learning & Noise Filtering

Signal Learning tracks how Ahmed engages with each flag pattern over time and uses that history to suppress recurring noise.

### `SignalLearning` tab structure

Headers: `Key Pattern | Total Seen | Acknowledged | Snoozed | Resolved | Expired/Ignored | Last Seen | Score | Suppressed`

The `Key Pattern` column stores the output of `extractKeyPattern_()`, which strips date and sequence suffixes from full flag keys:

- `verizon_bill_20260315` → `verizon_bill`
- `task_neglect_TASK_0012` → `task_neglect`
- `ergonomic_break_1234` → `ergonomic_break`

This normalization is critical: it means all variants of a recurring pattern accumulate into one row rather than spreading across dozens.

### Score formula

```
score = 100 - (snoozed × 20) - (expired × 15) + (ack × 10) + (resolved × 25)
score = clamp(0, 100)
```

- **Snoozed (-20)**: User acknowledged the flag was surfaced but chose to defer — mild negative signal.
- **Expired/Ignored (-15)**: Flag sat open > 30 days with no action — silent dismissal.
- **Acknowledged (+10)**: User read it and marked it done — mild positive signal.
- **Resolved (+25)**: User took action and resolved the underlying issue — strong positive signal.

### Suppression threshold

A pattern is marked `Suppressed = Yes` when: `score < 25` AND `total_seen >= 5`.

The minimum sightings gate (5) prevents suppression from triggering on a pattern that was snoozed once or twice early on. The score threshold (25) means a pattern needs to be consistently ignored or snoozed multiple times before VERA stops surfacing it.

### How suppression flows into the pipeline

1. **Step 0d** (`getSuppressedKeyPatterns_()`) reads the `SignalLearning` tab at the start of `nightlyRun()` and returns all suppressed patterns.
2. The array is passed into `generateFlags()` → `buildPrompt()` → injected as a "SUPPRESSED TOPICS" block: *"user has consistently indicated these are not useful — do NOT generate flags for them."*
3. **Anticipator rules** (Reminders.js) check `getSuppressedKeyPatterns_()` before sending nudges for bills due, packing reminders, goal check-ins, and home service items.

### Feedback loop

- `recordFlagsGenerated_(flagKeys)` is called after each `writeFlags()` call — increments `Total Seen` for each pattern key.
- `recordExpiredFlags_()` is called each night (Step 0e) — increments `Expired/Ignored` for flags open > 30 days.
- `recordFlagOutcome_(flagKey, outcome)` is called when a user acknowledges, snoozes, or resolves a flag via the dashboard or Slack.

`getTopSignalPatterns_(n)` returns the top N patterns by score (highest first). These are injected into the Claude prompt as "SIGNAL LEARNING — what Ahmed engages with most" to bias flag priority toward topics he cares about.

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## 5. Cross-Domain Pattern Recognition

`PatternRecognition.js` implements meta-intelligence: compound signals that span multiple VERA data domains. Single-domain flags (one overdue task, one missed gym session) are left to their individual checkers. The Pattern Recognition engine only fires when two or more domains signal simultaneously.

### `buildCrossDomainSnapshot_()`

Assembles a lightweight signal object from existing VERA data functions. Every data read is wrapped in its own `try/catch`: if a domain is unavailable (e.g. Gym Log tab empty), that field is set to `null` and any pattern requiring it is skipped gracefully. The snapshot includes:

| Field | Source |
|---|---|
| `overdueTaskCount` | `getOpenTasks()` — tasks where `isOverdue = true` |

| Field | Source |
|---|---|
| `neglectedTaskCount` | `getOpenTasks()` — tasks where `ageInDays >= 14` |
| `calendarEventsThisWeek` | `getUpcomingEvents()` — events in next 7 days |
| `activeGoals` | `getGoals_()` — goals with status `'Doing'` |
| `gymSessionsThisWeek` | `getGymLog_()` — sessions attended in last 7 days |
| `gymTargetPerWeek` | Config key `gym_sessions_per_week` (default 3) |
| `overdueHealthAppts` | `getHealthAppointments_()` — `daysUntil < 0` and `intervalMonths != 0` |
| `activeHighFlags` | Direct Flags tab read — unacknowledged, unresolved, urgency = `'High'` |
| `pacingMode` | `isInPacingMode_()` |
| `vacationMode` | `isInVacationMode_()` |
| `intensityLevel` | `computeIntensitySignal_()` — see Section 6 |
| `takeoutRatio` | `getMealPlanHistory_(2)` — takeout/eating-out fraction of meals in past 7 days |

## `evaluateCrossPatterns_(snap)` — the 7 patterns

All patterns null-check their required fields before evaluating. A pattern whose required fields are null is silently skipped.

**Pattern 1: High-Stress Compound** (`cross_high_stress_compound`, urgency: High)

- Trigger: `overdueTaskCount >= 3` AND `activeHighFlags >= 2` AND `gymSessionsThisWeek === 0`
- Rationale: tasks are piling up, high-urgency flags are unresolved, and the user isn't getting physical release — a compounding stress signal.

**Pattern 2: Goal–Behaviour Drift** (`cross_goal_behaviour_drift`, urgency: Medium)

- Trigger: `activeGoals >= 2` AND `gymSessionsThisWeek < gymTargetPerWeek * 0.5` AND `overdueTaskCount >= 2`
- Rationale: stated goals are active but execution is stalling on both fitness and tasks — declared intent is diverging from actual behaviour.

**Pattern 3: Social/Calendar Gap** (`cross_social_calendar_gap`, urgency: Low)

- Trigger: `calendarEventsThisWeek <= 1` AND `pacingMode === false` AND `vacationMode === false`
- Rationale: an unusually empty week that is not intentional (not pacing mode, not vacation) — may indicate social drift or missed opportunity for connection.

**Pattern 4: Overload + Pacing Mismatch** (`cross_overload_no_pacing`, urgency: Medium)

- Trigger: `intensityLevel === 'high'` AND `pacingMode === false` AND `activeHighFlags >= 2` AND `overdueTaskCount >= 3`

• Rationale: the system signals high load but the user hasn't activated pacing mode — they are running hard without a buffer.

**Pattern 5: Meal Chaos** (`cross_meal_chaos`, urgency: Low)

• Trigger: `takeoutRatio >= 0.7` AND `overdueTaskCount >= 2`

• Rationale: mostly takeout combined with task pile-up is a compound signal of a chaotic life period, not just a food preference.

**Pattern 6: Backlog Accumulation** (`cross_backlog_accumulation`, urgency: Medium)

• Trigger: `neglectedTaskCount >= 5` AND `calendarEventsThisWeek >= 3`

• Rationale: the calendar is full but tasks aged 14+ days are building up — the user is attending meetings but not processing their backlog.

**Pattern 7: Health Neglect Compound** (`cross_health_neglect`, urgency: Medium)

• Trigger: `overdueHealthAppts >= 2` AND `gymSessionsThisWeek === 0`

• Rationale: multiple overdue appointments combined with no gym sessions this week is a compound health-neglect signal.

### Deduplication and rate limiting

Candidates are sorted High → Medium → Low. Before writing, the function reads all existing `Pattern Recognition` source flags and builds a `suppressedKeys` map: a pattern key is suppressed if it has an open (unresolved) flag, or if it was resolved within the last `pattern_dedup_days` (default: 7) days. A maximum of `pattern_max_flags` (default: 2) new flags are written per run. Both values are overridable via Config tab keys `pattern_max_flags` and `pattern_dedup_days`.

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## 6. Pacing & Capacity Mode

### `computeIntensitySignal_(activeFlags, tasks, events)`

Defined in `WeekendPlanner.js` but used by both the Weekend Planner and `PatternRecognition.js`. Evaluates three thresholds:

• Active unresolved flags >= 4 → threshold breach

• Overdue tasks >= 2 → threshold breach

• Meetings in the next 5 days (non-all-day events) >= 8 → threshold breach

`level = 'high'` if 2+ thresholds breached, `'medium'` if 1 threshold breached, `'low'` otherwise.

Returns `{ level, activeFlagCount, overdueTaskCount, meetingCount }`.

### `isInPacingMode_()`

Fast read from the `VERA_STATE_PACING` Script Property. Reads `PACING_MODE_ACTIVE`. Auto-expires: if `today > PACING_MODE_ENDS` date, sets `PACING_MODE_ACTIVE = 'false'` and returns false. Called from `PatternRecognition.js`, `Fitness.js`, and `Reminders.js`.

### `isInVacationMode_()`

Returns `awayStatus_().on` (`Away.js`): true while Away mode is on, from a trip found by `checkVacationMode_()` (which runs inside `checkPacing_()` at the start of each nightly run) or from the manual switch, and not ended early.

## Away mode (`Away.js`, #246)

**Away mode** (`Away.js`, #246) is on while you're on a trip. VERA turns it on by itself each night when today falls within a trip in the Itinerary tab that isn't Victoria-only. You can also turn it on by hand from Home (✍■ Away mode…), for a trip VERA can't see, and end it early the same way. While it's on:

- Flags are still written, but nothing escalates.
- Each notification follows its **While away** setting (System › Notifications): **Send**, **Hold** (saved and counted for the catch-up), or **Send if due** (sent only when its deadline falls before you're back). The defaults: travel messages, ARIA Book now, the system logs and the watchdog send; High flags, bills, contracts and coupons send if due; everything else is held. Config `notif_<key>_while_away` overrides one.
- The morning email becomes the short **away edition**: today's plan, your flights, and only what got through. A flag gets through when it is High, comes from a source with a deadline (bills, contracts, coupons, card perks, travel documents), and its **Due** date (Flags tab column L) falls before you're back.
- Home shows a banner with what's held, and the capacity mode for trip days is `busy`.
- Fitness consistency checks, goal check-ins and pacing escalation are paused, as before.
- When Away mode ends (a trip of yours ends, the switch's date passes, or you tap "End away mode"), the first morning email after starts with **While you were away** (#244), once: the flags raised in those days (the 5 most urgent open ones listed, the rest counted), the tasks that went overdue (up to 5), what was held (one line, by kind), and the mail waiting. A part with nothing in it is left out, and with nothing at all there's no section. On a morning with no flags, when the briefing isn't sent, it goes on its own.

State lives in `VERA_STATE_PACING`: `AWAY_MANUAL`, `AWAY_UNTIL` and `AWAY_TRIP` for the switch; `AWAY_ENDED_TRIP` when a trip was ended early (cleared when the trip ends); `AWAY_STARTED`; `AWAY_HELD` (`{notification key: count}`); `AWAY_BACK`, the ended period waiting for the catch-up. `awayCloseIfEnded_` records it (from `checkVacationMode_` when a trip ends, from the switch, and at the start of `morningNudge` for a switch whose date has passed); `awayBackSection_` builds the box from the Flags tab, the open tasks and the mail counter; `morningNudge` clears it after the briefing is sent, or `sendAwayBackAlone_` sends it alone on a no-flags morning. The pure deciders are `awayDecide_`, `awayHoldsDecide_`, `awayFlagGetsThrough_`, `awayHeldSummary_` and `awayEditionHtml_`. Endpoints: `away_status` and `set_notif_while_away` (GET), `away_set` (POST `{ on, until, trip }`).

## `checkVacationMode_()` — vacation detection logic

Reads the `Itinerary` tab to build a date-range map (`tripKey → { min, max }`). Reads `TripMeta` to build a traveler map (`tripKey → traveler string`). Finds the first active trip where today falls within the date span and the trip is not Victoria-only (a trip is Victoria-only if the traveler string contains `'victoria'` and does not contain `'ahmed'`). Writes `VACATION_MODE_ACTIVE`, `VACATION_MODE_ENDS`, and `VACATION_TRIP_NAME` to `VERA_STATE_PACING`; on

a trip's first day it records `AWAY_STARTED` and clears `AWAY_HELD` (unless the switch already turned Away mode on). Fires a Memory Log event on first detection and on transition back to inactive.

### `checkMissRate_()` — the pacing offer loop

Scores 4 domains for misses in the last 48 hours:

1. **Gym**: any scheduled session today or yesterday with `attended != 'Yes'`
2. **Tasks**: any Open task with a due date of today or yesterday
3. **Flags**: `>= pacing_flag_threshold` (default 3) unacknowledged Medium/High flags open
4. **Chores**: any chore whose last `Checked At` + cadence interval has elapsed

If 2+ domains hit:

• **First detection**: Writes a Medium flag offering to activate pacing mode. Stores `PACING_FIRST_DETECTED` (timestamp) and `PACING_OFFER_FLAG_KEY` in Script Properties.

• **48 hours later (still 2+ domains hit, offer flag unacknowledged)**: Auto-activates pacing mode. Defers all upcoming non-recurring open tasks with a future due date to next Monday. Sets `PACING_MODE_ACTIVE = 'true'` and `PACING_MODE_ENDS = today + pacing_mode_days` (default 7 days). Writes a Low flag confirming activation.

Pacing mode effects: Pattern 4 (Overload+Pacing Mismatch) and Pattern 1 (High-Stress Compound) suppress their flags while pacing mode is active. The Anticipator's hydration, ergonomic, and mobility rules still fire — only contextually inappropriate nudges are suppressed.

---

## 7. Anticipator — Reminder Engine

`hourlyCheck()` fires every hour via the Apps Script `everyHours(1)` trigger installed by `setupTriggers()`. Apps Script does not support conditional hour windows in trigger configuration — the function itself contains all guard logic.

### `runAnticipatorRules_(now, hour, isWeekday, cfg)`

Runs 8 rules in sequence. Each rule is wrapped in its own `try/catch` — a failure in one rule does not block the others.

| Rule | When it fires | Cooldown |
|---|---|---|
| `checkErgonomicBreak_` | Weekdays 9am–6pm | 55 min (Slack) / 180 min (email) |
| `checkHydration_` | Weekdays 8am–6pm | 110 min (Slack) / 180 min (email) |
| `checkCalendarOpportunity_` | Weekdays 9am–5pm | 90 min (Slack) / 180 min (email), once per date+hour key |
| `checkEveningMobility_` | Configured hour (default 8pm) | 1440 min (once per day) |
| `checkBillsDue_` | Daily 8–9am | 1440 min per bill per day |
| `checkTripPackingReminder_` | Daily 8–9am | 1440 min per trip per day |

| Rule | When it fires | Cooldown |
|---|---|---|
| `checkGoalCheckin_` | Mondays at `weekend_planner_hour` (default 8am) | 10080 min (once per week) |
| `checkHomeServiceDue_` | Daily 8–9am | 1440 min per item per day |

### `Reminders Memory` tab (`TABS.REMINDERS_MEMORY`)

Every sent nudge is logged to this tab with columns: `Rule Key | Sent At | Message` (first 100 chars). `wasRecentlySent_(ruleKey, cooldownMinutes)` scans from the bottom (most recent first) for the most recent entry matching the rule key. If the elapsed time since that entry is less than `cooldownMinutes`, the rule is blocked.

### Suppression integration

`checkBillsDue_()`, `checkTripPackingReminder_()`, `checkGoalCheckin_()`, and `checkHomeServiceDue_()` each call `getSuppressedKeyPatterns_()` before firing. If the pattern key for that bill, trip, or home item is suppressed, the nudge is skipped entirely.

### `sendNudge_(ruleKey, subject, message)`

Delivery logic:

1. If `isSlackConfigured_()` is true (i.e. `SLACK_BOT_TOKEN` is set in Script Properties): sends to `#vera-notifications`.
2. Else: sends a plain-text email via `MailApp.sendEmail()` to `CONFIG.MORNING_NUDGE_EMAIL`.

Always calls `markSent_(ruleKey, message)` after delivery to record the send in `Reminders Memory`. Also calls `sendSlackLog_()` to post a delivery confirmation to `#vera-logs`.

### Weekend Planner integration

`hourlyCheck()` contains an additional gate: `if (day === 3 && hour === cfg['weekend_planner_hour'])`, which calls `runWeekendPlanner_()`. This fires every Wednesday at the configured hour (default 8am). The planner uses a 9000-minute cooldown in `Reminders Memory` (approximately 6.25 days) to prevent double-sends.

`runWeekendPlanner_()` assembles clear weekend windows (via `findClearWindows_()` filtered to windows that touch a Friday or Monday), computes the intensity signal, reads goals, the Shared Interest Ledger, and PTO stats, then calls Claude to generate a "Weekend Decision Memo" with three archetypes:

- **THE EXTENSION**: goal-anchored activity
- **THE CONTRAST**: rest/recharge (weighted heavier if intensity is high)
- **THE PROTOTYPE**: new experience not in the Interest Ledger

The memo is delivered via `sendNudge_()` and also written as an all-day Google Calendar event on the upcoming Saturday in the `pto_vera_calendar` (default: `'Vera'` calendar).

---

# 8. Chat System Architecture

## Context assembly

Every chat turn in `Chat.js` assembles a fresh context object via `buildChatContext_()`. The context includes:

- Active flags (unresolved, unacknowledged) from the Flags tab
- Open tasks from `getOpenTasks()`
- Upcoming events from `getUpcomingEvents()`
- Life summaries from `getSummaries()`
- Bills from the Bills tab
- Home items from the Home Items tab
- Goals from `getGoals_()`
- Financial goals
- Projects
- Travel context (upcoming trips, packing items by trip key, itinerary items by trip key)
- Career position, goals, progression, development, wins, network
- And more domain-specific data depending on the chat turn context

This context is assembled fresh on every turn — there is no persistent in-memory state between chat calls (Apps Script does not support persistent processes).

### `buildChatSystemPrompt_(context)`

The system prompt injects the assembled context as structured sections for Claude to reason over. It also injects:

- Proactive insights from `computeProactiveInsights_(context)` — surfaced as "VERA NOTICES" in the prompt
- Source routing identity (Ahmed vs Victoria vs Slack)
- Persona definition and action capabilities

## Proactive insights — `computeProactiveInsights_(context)`

Pre-computes from context data (no additional API calls) and returns a formatted `[HIGH/MEDIUM/LOW]` notice string. Sources checked:

1. Tasks overdue > 3 days (by name, sorted by staleness, top 3)
2. Bills due within 7 days and unpaid (urgency: `<= 2 days = HIGH`, `<= 5 = MEDIUM`, `<= 7 = LOW`)
3. Upcoming trips within 14 days: empty packing list (`HIGH` if within 7 days), unpacked items (`MEDIUM`), empty itinerary (`MEDIUM`)
4. Home items with service overdue (`HIGH`) or due within 7 days (`LOW`)
5. Projects with overdue tasks (`MEDIUM`)

These are injected into the system prompt so Claude can surface them proactively at the start of a conversation or when contextually relevant.

### Source routing

The `body.source` field determines how Claude identifies the conversation participant:

- `'dashboard'`: Ahmed (primary user)
- `'dashboard-lite'`: Victoria (simplified dashboard)
- `'vera-chat'`: Slack `#vera-chat` channel — identity derived from Slack user ID via `getSlackUserName_()`

### Action dispatch

Chat.js contains a large `dispatchAction_()` function that handles Claude's structured action responses. Claude returns a JSON action object (`{ action, params }`) alongside its conversational reply. Supported actions include (but are not limited to): `add_task`, `complete_task`, `resolve_flag`, `snooze_flag`, `acknowledge_flag`, `update_goal`, `add_trip_item`, `add_packing_item`, `log_gym_session`, `add_bill`, `add_home_item`, `update_project_task`, `record_purchase`, and many more. Each action modifies the appropriate sheet tab directly.

### Multi-turn context

Chat history is stored per session in Script Properties under the key `CHAT_HISTORY_{sessionId}` as a JSON-serialised array of `{ role, content }` messages. The `CHAT_MAX_EXCHANGES` constant (default: 10) limits the stored back-and-forth to 20 messages (10 pairs). Older messages are trimmed from the front of the array.

### Web search

When `VERA_SEARCH_API_KEY` is configured in Script Properties, the `WEB_SEARCH_TOOL_` is appended to the Claude API call's `tools` array. Claude can invoke it for real-time or location-specific queries. A PII scrub (email, phone, SSN regex) runs on every query before it leaves Apps Script. The engine is configurable via `VERA_SEARCH_ENGINE` Script Property (`'serper'` or `'tavily'`, default `'serper'`).

---

## 9. Slack Integration Architecture

### Inbound (user → VERA)

Slack sends `POST` requests to the Apps Script web app `doPost()` endpoint via the Events API. Two payload shapes arrive:

1. **JSON body** (`body.type === 'event_callback'`): routed to `handleSlackEvent_()` in `Slack.js`
2. **Form-encoded body** (button interactions, slash commands): routed to `handleSlackFormPost_()` in `Slack.js`

Apps Script does not reliably decode form-encoded POST bodies into `e.parameter`. `handleSlackFormPost_()` falls back to manually URL-decoding `e.postData.contents` when `e.parameter` is empty.

**Bot message filtering**: `handleSlackEvent_()` checks `event.bot_id` and `event.subtype === 'bot_message'` before processing. VERA's own outbound messages arrive back via the Events API and must be

ignored to prevent echo loops.

**Slack user allowlist**: `getSlackAllowedUserIds_()` reads `SLACK_ALLOWED_USER_IDS` from Script Properties (comma-separated Slack user IDs). Only allowlisted users can trigger VERA via Slack. Unknown users receive a silent rejection.

**Async queuing**: Chat messages from `#vera-chat` are queued in `CacheService` and processed asynchronously. This is required to respond within Slack's 3-second acknowledgement deadline — the Apps Script execution (which involves calling Claude) can take 10–30 seconds.

## Outbound (VERA → user)

Three send functions handle outbound messages:

- `sendSlackMessage_(channelId, text, blocks, threadTs)`: Generic send to any channel. Accepts optional Block Kit `blocks` array and optional `threadTs` for threading.
- `sendSlackNotification_(text, blocks, urgency)`: Routes to `#vera-notifications` (Config `slack_notifications_channel_id`). For `urgency === 'High'`, prepends `<@AHMED_USER_ID>` to trigger a Slack mention.
- `sendSlackLog_(text)`: Plain text to `#vera-logs` (Config `slack_logs_channel_id`). Called by `nightlyRun()`, reminder delivery, pattern recognition, travel pipeline, and monthly review.

## Block Kit

`buildFlagBlocks_(flag)` assembles a three-block structure for each flag:

1. `section` block with urgency emoji + bold flag title + italic reason (truncated to 200 chars)
2. `actions` block with two buttons: `acknowledge_flag` (action_id) and `snooze_flag` (action_id), each carrying the flag ID as the `value`
3. `divider` block

When a user clicks Acknowledge or Snooze, Slack sends an interaction payload (form-encoded JSON) to `doPost()`. `handleSlackInteraction_()` parses the action ID and value, updates the flag in the sheet, and calls `sendSlackResponse_(responseUrl, confirmText, null, true)` with `replace_original: true` to replace the button message with a confirmation.

The morning nudge uses `buildMorningNudgeBlocks_()` which groups flags by urgency (High → Medium → Low) and includes a capacity mode ticker and a "held from yesterday" count.

## Channel routing

| Channel | Script Property key | Purpose |
|---|---|---|
| `#vera-chat` | Config `slack_chat_channel_id` | Bidirectional conversation (Ahmed + Victoria) |
| `#vera-notifications` | Config `slack_notifications_channel_id` | Flag alerts with Acknowledge/Snooze buttons |

| Channel | Script Property key | Purpose |
|---|---|---|
| `#vera-logs` | Config `slack_logs_channel_id` | System event log — nightly run, feature fires, errors |

## 10. Travel Pipeline

### Trip detection

Trips are identified by the `Itinerary` tab. Each row has a `Trip Key` column with the format `YYYY-MM-DD|Trip Label` (e.g. `2026-06-15|Nashville Weekend`). The departure date is the `YYYY-MM-DD` prefix. The `TripMeta` tab stores per-trip context: a traveler field identifies which trips are Ahmed-only, Victoria-only, or joint. Victoria-only trips are excluded from `checkVacationMode_()` and the pacing system.

### Email parser — `runEmailScan_()`

Runs every 30 minutes via the `runEmailScan_` trigger (only when `email_parser_enabled = 'true'` in Config). The process:

1. Runs a broad Gmail search via `buildTravelSearchQuery_()` — returns up to 50 threads
2. Deduplicates against the `Processed Emails` tab (`isAlreadyProcessed_(messageId)`)
3. Batches new candidates (up to `BATCH_SIZE = 20`) to Claude with a confidence scoring prompt
4. Per-email response: `confidence >= HIGH_CONF (0.85)` → auto-process (write/enrich itinerary row); `LOW_CONF (0.60) <= confidence < HIGH_CONF` → hold and write a flag for manual confirmation; `confidence < LOW_CONF` → discard silently
5. Marks every processed email in `Processed Emails` tab with outcome and mode

The email body is truncated to `MAX_SIGNAL_CHARS = 2500` characters before sending to Claude. The warning in `setupTriggers()` notes that with emails per day up to the 50-thread search limit and 30-minute intervals, this could generate up to 144 Claude API calls per day — the config key `email_parser_enabled` defaults to `'false'`.

### Pre-trip briefing — `checkPreTripBriefings_()`

Reads the Itinerary tab, finds trips departing within `pretrip_briefing_hours` (default: 48 hours), and assembles a `buildPreTripBriefingFlag_()` for each. The briefing flag is written as `urgency: 'High'` with source `'Trip Briefing'`. Deduplication via `writeFlags()` key fingerprinting ensures one briefing per trip, ever — the key includes the trip key, so it fires once and never again even if the nightly run repeats. Logged to `#vera-logs`: `:airplane: Pre-trip briefing written — Nashville Weekend`.

### Flight status monitor — `checkFlightStatuses_()`

Polls the AviationStack API every 15 minutes via the `checkFlightStatuses_` trigger. Active only for flights within 24 hours of departure. Polling intervals are rate-limited:

- Departure > 6h away (in the 6–24h window): every 3 hours
- Departure 1–6h away: every 60 minutes
- Departure < 1h away: every 15 minutes

Status is stored in: (a) the Itinerary tab row's `Metadata` column (JSON with `flight_status` key) for sheet-sourced flights; (b) the `FLIGHT_STATUS_CACHE` row of the hidden VERA State tab (keyed by `CAL-xxx` IDs) for calendar-sourced flights.

When a 429 response is received from AviationStack: a monthly quota error backs off for 30 days; a rate-limit error backs off for 2 hours. The backoff window is stored in `AVIATIONSTACK_BACKOFF_UNTIL` Script Property and checked at the top of every poll.

### Post-trip capture — `checkPostTripCapture_()`

Fires `posttrip_capture_delay_days` (default: 1) after a trip's end date. Writes a flag (source: `'Trip'`) prompting Ahmed to log memories, expenses, and notes. Gated by `posttrip_capture_enabled = 'true'`.

---

## 11. PTO & Clear Windows

### PTO snapshot — `writePTOSnapshot_()`

Reads Ahmed's work calendar (configured via `pto_calendar_name` Config key, default `'Verizon Calendar'`). Classifies all-day events:

- Title contains `'Vacation'` → vacation day (pool: `pto_vacation_days`, default 20)
- Title contains `'PTO'` (all-day) → 8 personal hours (pool: `pto_personal_hours`, default 48)
- Title contains `'PTO'` (timed) → actual event duration in hours
- Title matches `pto_holiday_keywords` (default: `'Day,Holiday,Floating,Closure'`) → company holiday (not counted against PTO pools)
- Title matches `pto_ignore_keywords` (default: `'Pay Day'`) → skip entirely

Computes: days used, days remaining, ideal burn-rate pace (remaining days / remaining weeks in year), pace gap (actual vs ideal). Writes `[AUTO]` rows to the Summaries tab covering vacation balance, personal time balance, and pace status. Returns `ptoStats` object passed to Monthly Review.

### Clear windows — `findClearWindows_()`

`findClearWindows_(gapCalendars, today, lookAheadDays, minWorkdays)` scans the next `lookAheadDays` days and finds contiguous stretches of `minWorkdays` or more clear workdays. "Clear" means no events on calendars defined by `pto_gap_calendars` Config key (default: `'Verizon Calendar,AE&VV - Our Joint Chaos'`).

Used by `getWeekendWindows_()` in `WeekendPlanner.js` (filtered to windows touching a Friday or Monday) and by the Health Tracker to suggest appointment booking windows.

### 3-2-1 Framework

The PTO system surfaces a 3-2-1 planning framework in the dashboard PTO tab:

- **3 months out**: Plan big trips and time-off blocks
- **2 weeks out**: Confirm logistics, book details
- **1 day out**: Verify everything is ready

The `ptoSummaryForClaude_()` function formats PTO data for injection into the Claude prompt, including 3-2-1 status indicators and a `paceStatus` field (`behind`, `on_track`, `ahead`).

---

## 12. Summaries Architecture

### Two-tab architecture

**Metrics tab** (`TABS.METRICS`): VERA's self-monitoring. Written by `writeMetrics_(ss)`. Contains task counts (open, overdue, neglected), calendar event counts (today, this week), and flag counts (total, by urgency). Fully automatic — no user configuration needed. Rows are `[AUTO]`-prefixed and rewritten every night.

**Summaries tab** (`TABS.SUMMARIES`): External life intelligence. Written by `writeSummaries_(ss)`. Populated from external Google Sheets via `summary_sheet:SourceName` Config rows. Also receives `[AUTO]` rows from PTO snapshot, fitness data, and other module writes.

### AUTO rows

Any row whose `Source` column starts with `[AUTO]` is owned by the script. The `clearAutoRows_(sheet, tabName)` function:

1. Reads all rows in the sheet
2. Collects row indices where `Source` starts with `'[AUTO]'`
3. Deletes them from bottom to top (to preserve row indices during deletion)

After clearing, new `[AUTO]` rows are inserted before row 2 (just after the header) and styled with `background: '#eef2ff'` (subtle blue tint) and `fontColor: '#333333'`. Manual rows (without `[AUTO]` prefix) are never touched.

### External sheet hooks

Config tab syntax:

```
Setting                          | Value
summary_sheet:SimpleAssTracker   | SHEET_ID|TabName|CellRef|metric_name
```

For example:

```
summary_sheet:SimpleAssTracker   | 1aBc...XyZ|Budget|B2|checking_balance
summary_sheet:SimpleAssTracker   | 1aBc...XyZ|Budget|B5|groceries_actual_vs_budget
```

```
summary_sheet:Fitness          | 1xYz...AbC|Log|D4|gym_sessions_this_week
```

`writeSummaries_()` reads all Config rows with keys starting with `summary_sheet:`. For each, it opens the external sheet via `SpreadsheetApp.openById(sheetId)`, reads the specified cell, and writes an `[AUTO]` row to the Summaries tab: `source = '[AUTO] SourceName'`, `metric = metric_name`, `value = cellValue`, `asOf = today`.

The external sheet must be shared (at minimum viewer access) with the Google account running the script.

### `getSummaries()`

Reads both the Metrics tab and the Summaries tab and returns a merged array of `{ source, metric, value, asOf }` objects. This merged array is what `generateFlags()` and chat context assembly receive — they do not distinguish between the two tabs.

---

## 13. Monthly Life Review

### Trigger

`checkMonthlyReview_(ptoStats)` is called from `nightlyRun()` Step 0o. The function immediately returns if `today.getDate() !== 1` — it only runs on the 1st of each month. A second guard checks whether a review with the key `monthly_review_YYYYMM` (for the prior month) already exists anywhere in the Flags tab (any status) — if found, it returns without re-generating. This dedup ensures the review fires exactly once per month even if `nightlyRun()` somehow triggers multiple times on the 1st.

The Config key `monthly_review_enabled` (default `'true'`) gates the entire feature.

### Assembly — `buildMonthlyReview_(label, priorMonth, ptoStats)`

Assembles the review by calling independent section builders, each wrapped in `try/catch`:

- **Goals**: Current-year goals grouped by status (Done, Doing, Stalled, Backlog)
- **Tasks**: Open task count, overdue count, most neglected tasks by age
- **Finance**: Reads spending data from the Summaries tab — `[AUTO]` rows with finance/SAT source
- **PTO**: Uses the `ptoStats` object passed from Step 0b (avoids re-reading the calendar)
- **Travel**: Reads the Itinerary tab for trips whose date range overlaps the prior month
- **Flags**: Counts flags generated in the prior month, resolved vs still-open
- **Carry Forward**: A single Claude call asking for one insight to carry forward from the month's data

### Output

Three outputs are produced:

1. **Flag**: Written via `writeFlags()` with `source: 'Monthly Review'`, `urgency: 'Low'`, `key: 'monthly_review_YYYYMM'`. The full review text goes in the `reason` field.
2. **Archive**: Appended to the `Monthly Reviews` tab (headers: `Month Key | Month Label | Generated Date | Review`) as an append-only record. This tab is never modified by VERA — only appended.
3. **Email**: Sent via `sendMonthlyReviewEmail_(label, reviewText)` as a styled HTML email.
4. **Slack log**: `sendSlackLog_('■ Monthly review written — March 2026')`.

---

## 14. Configuration Architecture

### Config tab (runtime-tunable)

The `Config` tab in the Life OS sheet contains key-value pairs read by `getConfigValues()` on every execution:

```
function getConfigValues() {
  if (_configCache_) return _configCache_;  // per-execution cache
  var ss    = getSpreadsheet();
  var sheet = ss.getSheetByName(TABS.CONFIG);
  // ... reads all rows into an object ...
  _configCache_ = cfg;
  return cfg;
}
```

`_configCache_` is a module-level variable declared as `var _configCache_ = null` in `Code.js`. It resets automatically at the start of each Apps Script execution (there is no persistent memory between executions). Within a single execution (e.g. a nightly run), `getConfigValues()` is called 4–6 times — the cache prevents the Config tab from being re-read on every sub-function call.

Config keys that change behaviour have sensible defaults hard-coded as fallback strings (e.g. `cfg['calendar_days_ahead'] || '7'`). This means the system works out of the box with an empty Config tab; users only add rows to override defaults.

Config tab keys used across the system include:

| Key | Default | Module | Purpose |
|---|---|---|---|
| `calendar_days_ahead` | `'7'` | Calendar.js | Look-ahead window for events |
| `task_age_threshold_days` | `'7'` | Tasks.js | Age before a task is "neglected" |
| `max_flags_per_night` | `'8'` | Claude.js (`CONFIG.MAX_FLAGS`) | Cap on Claude flag output |
| `skip_calendars` | `'Holidays in United States'` | Calendar.js | Calendars to exclude entirely |

| Key | Default | Module | Purpose |
|---|---|---|---|
| `calendar_label:CalName` | (none) | Calendar.js | Custom label for a calendar name |
| `pto_vacation_days` | `'20'` | PTO.js | Annual vacation allocation |
| `gym_sessions_per_week` | `'3'` | PatternRecognition.js, GymTracker.js | Weekly gym target |
| `reminders_enabled` | `'true'` | Reminders.js | Master switch for Anticipator |
| `explorer_enabled` | `'true'` | Reminders.js | Master switch for Explorer |
| `email_parser_enabled` | `'false'` | EmailParser.js | Master switch for inbox scanner |
| `pretrip_briefing_enabled` | `'true'` | PreTripBriefing.js | Enable/disable pre-trip flags |
| `monthly_review_enabled` | `'true'` | MonthlyReview.js | Enable/disable monthly review |
| `pattern_max_flags` | `'2'` | PatternRecognition.js | Max cross-domain flags per run |
| `pattern_dedup_days` | `'7'` | PatternRecognition.js | Dedup window for pattern flags |
| `pacing_enabled` | `'true'` | Pacing.js | Enable/disable pacing system |
| `pacing_flag_threshold` | `'3'` | Pacing.js | Unacked flags to trigger miss scoring |
| `summary_sheet:SourceName` | (none) | Summaries.js | External sheet hook definition |

## Script Properties (deployment secrets)

Read by `PropertiesService.getScriptProperties()`. These are set once in the Apps Script editor (Project Settings → Script Properties) and never committed to source control.

| Property | Used by | Purpose |
|---|---|---|
| `VERA_SHEET_ID` | `Code.js` (`CONFIG.SHEET_ID`) | The Life OS Google Sheet ID (needed before the Config tab can be read) |
| `CLAUDE_API_KEY` | `Claude.js` | Anthropic API key |
| `VERA_WEB_TOKEN` | `WebApp.js` | Token on every dashboard request |
| `SLACK_BOT_TOKEN` | `Slack.js` | Slack bot OAuth token |
| `SLACK_ALLOWED_USER_IDS` | `Slack.js` | Comma-separated allowlist of Slack user IDs (who may talk to VERA) |

| Property | Used by | Purpose |
|---|---|---|
| AVIATIONSTACK_KEY | FlightStatus.js | AviationStack API access key |
| WEATHER_API_KEY | Weather.js | OpenWeather API key |
| GOOGLE_STATIC_MAPS_API_KEY | TravelDayBriefing.js, TravelLegs.js | Google Maps key |
| VERA_SEARCH_API_KEY | Chat.js | Serper.dev or Tavily API key for web search |

The ids, links, channel and user ids, the email address and the search engine are **Config rows**, not Script Properties: morning_nudge_email, vera_dashboard_url, vera_logo_file_id, sat_sheet_id, transactions_sheet_id, address_book_sheet_id, life_plan_doc_id, shopping_list_doc_id, slack_chat_channel_id, slack_notifications_channel_id, slack_act_now_channel_id, slack_logs_channel_id, slack_ahmed_user_id, slack_victoria_user_id, vera_search_engine, neighborhood_hoa_url, and roster_link_aria for ARIA. veraSetting_ (Settings.js) reads the Config row first, then the old Script Property of the same name in capitals, so a deployment that hasn't moved still works. moveSettingsToConfig(), run once from the editor after a backup copy of the sheet, moves them: it fills only blank Config cells, reads each back before deleting the property, leaves a Config row that holds a different value alone, and logs names and counts, never values. The one-time "already done" markers live together in VERA_ONCE_MARKERS (veraOnceGet_/veraOnceSet_), and VERA_SETTINGS_COPY is a nightly copy of the settings used only when the Config tab can't be read. Each feature's runtime state is one record (veraStateGet_ / veraStateSet_, or the veraStateProps_ adapter): VERA_STATE_PACING (pacing and Away mode), VERA_STATE_PTO_BUFFER, VERA_STATE_MAIL, and VERA_STATE_CAPACITY_OVERRIDE apart from VERA_STATE_CAPACITY_INFERRED, so a nightly write never undoes your override. Caches and histories (FLIGHT_STATUS_CACHE, FLIGHT_DAYS, EXPLORER_HISTORY, WKND_PLAN_HISTORY) are rows in the hidden **VERA State** tab (veraStoreGet_ / veraStoreSet_), read once per run, falling back to the Script Property if the tab can't be written. A value not yet moved is still read from its old property, and moveSettingsToConfig() moves them all. What's left in the store is secrets and run stamps.

**Writes that two runs could make at once (#242).** VERA's triggers and the dashboard can run together, so a read-modify-write goes through veraLocked_ (Settings.js): the per-user lock, which

never nests with the script locks modules take themselves, and goes ahead (logging it) if it stays busy. Every Config row is written by `setConfigRow_` under that lock: the PTO values, notification switches and channels, the grocery stores, and dashboard settings. Habit and memory IDs come from `veraNextSeqId_`, the highest number in use plus one, minted and appended under the lock, so a deleted row or two writers at once can't repeat an ID. The SignalLearning counters update under the lock too.

## Decision rule for where a value lives

- **Secret or credential** (API key, sensitive token, who may talk to VERA) → Script Property. Never in the sheet, never in source control.
- **An id or link that isn't a secret** (a sheet, doc or file id, a Slack channel or user id, the dashboard link, an email address) → Config tab, read with `veraSetting_`. Never in source control either: the repo is public.
- **Value the user might want to tune** (thresholds, labels, calendar names, feature flags) → Config tab with a sensible default fallback in code.
- **Value that changes system behaviour** (feature on/off switches, interval days, target counts) → Config tab with a sensible default fallback. The fallback ensures the system works without any Config row being present for that key.
- **Value that is computed at runtime and needs to survive between executions** (pacing mode active/inactive, Away mode, AviationStack backoff window) → Script Properties as ephemeral state. These are not secrets but they need to persist across the 6-minute execution boundary.

## The Config tab's layout (November 2026)

`getConfigValues()` reads columns A and B only, keyed by the name in column A. Rows whose column A is blank are skipped, and a section title row is just a key with an empty value. So the tab can be laid out for people without the code noticing:

- **Columns:** A *Setting* · B *Value* (the only column VERA reads) · C *What it does* · D *Where it's used* (a `=HYPERLINK("#gid=…")` to the related tab).
- **Order:** a Contents block of links (rows with an empty column A), then system-wide sections (General, Notifications, Calendars, Linked files, Assistants), then one section per feature. Each section is a title row plus a collapsible row group.
- **Defaults made visible:** settings the code reads with an inline fallback (`cfg['x'] || 'default'`) are listed with that same value and "(default)" in column C, so changing them is a matter of editing B. A row whose value equals the code default behaves exactly like a missing row.
- **Writers** (`Groceries.js`, `PTO.js`, the notification channel route in `WebApp.js`, `ComingUp.js`) all find their row by key and write column B, or append; none depend on row positions.
- **Data tabs are different:** they are read by position (`getRange(2, 1, n, HEADERS.length)`), so columns there are never moved or inserted. Formatting (colours, banding, frozen rows, number formats on numeric cells, tab order and tab colours) is safe. `SheetStyle.js` (#235) holds that style as a script: run `styleVeraSheet()` from the Apps Script editor to re-apply it to every data tab (navy header, banding over the whole grid, a tab colour and order per area, money, wrap and clip by header name, Urgency and Status colours, resolved Flags greyed). It never writes a value or moves a column, leaves Config and Access as they are, replaces only its own colour rules on a re-run, and stops before the time limit (run it again to finish). Tabs VERA creates later get the banding and area colour from

`ensureSheet`. It never runs in the nightly run.

### Settings you're most likely to change

Change only column B. The full list of keys is in the appendix (§22).

| Section | Setting | What it does |
|---|---|---|
| General | `morning_nudge_time` | Hour the morning briefing goes out |
| General | `max_flags_per_night` | Most new flags a night |
| Notifications | `notif_<key>_channel`, `notif_<key>_enabled`, `notif_<key>_while_away` | Where each message goes, or off, and what it does while you're away: send, hold or due (System › Notifications edits them) |
| Calendars | `gap_calendars`, `skip_calendars` | Calendars checked for free time; calendars ignored |
| Travel | `home_airports`, `pretrip_briefing_hours` | Where you fly from; how early the pre-trip email goes |
| Travel | `victoria_email` | Victoria's address for her Travel Day email |
| PTO | `pto_vacation_days`, `pto_rollover_days`, `pto_year` | Balances for the year (the same keys with `victoria_` for Victoria) |
| Finances | `finance_skip_categories` | Categories left out of spending |
| Home | `shop_grocery_stores` | Which stores count as groceries |
| Fitness | `fitness_weekly_target` | Workouts a week you aim for |
| Life | `coming_up_days`, `home_address` | How far ahead Coming up looks; where leave-by times start |
| Assistants | see §21.6 | Links to WILLA, TESS and ARIA |

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## 15. Trip Identity, Emails & Readiness

**Trip registry (`Trips.js`).** Every trip gets an immutable `TRIP-…` ID in the *Trips* tab. Label and dates track the calendar; earlier keys (`startDate|label`) are kept in *Aliases*. Resolution is layered: calendar event ID, then alias, then label + date proximity (14 days), never a guess when two candidates tie. Every read of a trip-keyed row goes through `tripKeyMatches_` / `tripKeysFor_`, so rows written under an old key still belong to the trip after a date change. Daily latches (Travel Day email, pre-trip briefing) key on the Trip ID, so a moved trip never emails twice.

**Trip ID columns (`TripIds.js`, #239 phase 1).** Ten trip tabs carry a Trip ID as their last column: Itinerary, TripMeta, PackingItems, Trip Budget, Trip Gifts, Trip Outfits, Packing Reviews, Leaving Home Runs, TripRecommendations and Trip Decisions. Fixed-width writers fill it (`tripIdCell_`); `tripIdStampAll_` fills any blank one from its key each night,

right after legacy keys are adopted, and never overwrites an ID that is there. Run `backfillTripIds()` once by hand after the release, with a backup of the sheet first. Readers still match by key; switching them to the ID is phase 2.

**Who's travelling and Travel Day recipients (`TripEmails.js`).** TripMeta's *Traveler* column holds names from the traveller profiles (old one-word values such as `Both` still read). The Trips tab's *Email Recipients* holds the switched-on travellers' names and any added addresses, separated by `;`. `teResolve_` builds the To list: the owner always on trips he travels on; Victoria only when switched on, with her address from Config `victoria_email`; others from their profile email; added addresses as given; an added address that matches a traveller's (Victoria's own, say) switches that traveller on. Victoria's solo trips send no Travel Day email. Config `travel_companions` is retired.

**Every day of a trip (#237).** The Travel Day email goes out on each day of a trip, and knows which day it is: `travelDayKind_` makes it departure, return, transit (a flight mid-trip) or "during", with "Day n of N" in the subject and header on the days in between. `travelStayTonight_` finds tonight's hotel, arranged stay or cruise, and `travelDayArea_` where the day's stops are. The narrative prompt is told all three, so a port day is never written as a departure. The per-stop details lookup is told them too: on a cruise it gives directions from the pier with no parking, a meal with no address is looked for on board first, and a same-named place in another city or country comes back as not found. The "data is not live" notice leaves out flight data on a day with no flight.

**Traveller profiles (`TravelerProfiles.js`).** The *Traveler Items* tab holds one row per document, membership, preference or health item per person. The server keeps only the last 4 digits of any number. Held visas and ETAs mark a destination as covered in the documents check; a nightly pass reminds about passports, ETAs and trusted-traveller expiries ahead of trips (notification `travel_docs`).

**Readiness (`TripReadiness.js`).** `trdReadiness_` turns a trip, its itinerary metadata, documents, packing and prescriptions into rows per traveller: ready / to do / blocks (documents, member number on bookings, seat preference, lounge access by status or card, refills due before the trip ends).

## 16. Packing, Luggage & Leaving Home

- `Luggage.js` **(#230):** "Our bags" (size, carry-on/checked, empty weight), per-person bag advice from nights, weather, trip type and the list, about-weights against the fare's allowance, and which status or card waives the checked bag. Road trips get bag advice only.
- `LeavingHome.js` **(#231):** checklist templates per home, filtered by season, length and who stays; runs per trip or for ad-hoc "Heading out" time away. Ticks are shared and record who ticked. The night-before email carries the list; a morning check sends `leave_home` if items remain.
- `PackPlus.js` **(#234):** status per item (`To buy → Ready → In the bag`), quantities, who brings it, staples (`always_pack` profile items, offered after generating), copy from a similar past trip, outfits per day, a Shopping "Trip: …" buy list, a checkout tick-off, and a post-trip review that feeds the next list.

## 17. Finances (November 2026)

- **Budget (`Budget.js`):** reads the budget tracker's Tracker tab live and by label; works out plan vs actual and a What-if overlay without writing the tracker.
- **Bills & cashflow (`BillSchedule.js`):** the Bill Schedule tab (due day, account override, autopay, card) joined to the tracker's bills, and a joint-account timeline with the opening balance and transfer day.

- **Summary & check-in (`FinSummary.js`):** quarterly snapshots (*Finance Snapshots*), decisions with owners (*Check-in Decisions*), notes, and the `quarterly_checkin` reminder.
- **Groceries and receipt runs (`Groceries.js`, `GroceryRun.js`):** a receipt photo is read by Claude vision (store, date, lines, total); picked lines go to Purchase History (On Hand syncs from it), matching shopping-list items are ticked across stores, and the run is logged to *Grocery Runs* with its date.
- **Phone plan, Planned Purchases, planner Goals and Net worth** read their own sheets; the planner is read live and never written.

## 18. Places, Gifts, Life & Career

- **Places (`Places.js`, `BucketPlus.js`):** Countries (map/list/timeline, occasions) and the Bucket List with things to do, best months (cached per place in *Place Facts*), vacation-day cost, award points from Config `award_*` rows, and a long-weekend filter by direct flight time from `home_airports`.
- **Gifts (`TripGifts.js`):** kind, status, budget, history per person, ideas per destination (cached in *Gift Place Ideas*) and customs notes.
- **Life (`Life.js`, `LifeGrowth.js`, `Growth.js`, `ComingUp.js`, `QuickCapture.js`):** Explore, Growth (Now, Goals, Habits, Learning, Trials, Review), Coming up (calendar events by type with lead days from `coming_up_type:*`), and Quick capture, which classifies one line of text and files it with an Undo.
- **Career Credentials (`CareerCredentials.js`):** certifications and degrees with expiry and renewal credits; reminders at 90/30/7 days and behind-pace (`career_cert_expiry`); shared with the job agent.

## 19. Family Planning & the Cycle Tracker

`FamilyPlanning.js` holds one shared plan (items, leave blocks, policy) by phase. `Cycle.js` adds a *Cycle Log* (start, end, who logged). Pure functions compute stats (average of the last six gaps, outliers ignored), today's view (cycle day, mood note from `cycle_mood_days`, next period, fertile window and peak days) and due notices. Notices are sent only while the phase is Trying for the fertile window, once per key, through `cycle_reminders` to a channel chosen in System › Notifications. Nothing goes on the shared flag list.

## 20. Dashboards, Navigation & Notifications

- **Two dashboards:** `docs/index.html` (built from `docs/app.js` by `docs/build.js`, React inlined) and `docs/dashboard-lite.html` (Victoria's, compiled in the browser). Shared blocks are copied byte-for-byte between them and tests enforce it.
- **Tile navigation:** each area groups its sub-tabs into tiles (`VN_GROUPS`, `FIN_GROUPS`, `HF_GROUPS`, `LF_GROUPS`, `SYS_SUBS`). The lite dashboard has eight tabs: Today, Chat, Shopping, Home, Travel, Money, Family, Projects.
- **Notifications:** every message is listed in `NOTIF_REGISTRY` (`Code.js`) with a default channel; Config `notif_<key>_channel`, `notif_<key>_enabled` and `notif_<key>_while_away` override it, and System › Notifications edits them.

# 21. Your Assistants and How They Connect

VERA is one of six assistants. Each runs on its own and keeps its own data; this section records every place two of them touch, who writes what, and what breaks when a link is missing. The dashboard's Home roster ("Your assistants") links each one from a Config row `roster_link_<name>` (a link or a bare sheet ID, read at run time, never written in the repo).

| Assistant | What it is | Where it runs | Link with VERA |
|---|---|---|---|
| **PIES** (Personal Intelligence & Execution System) | Work signal turned into prioritised actions; the work Done-Log | The work machine, in Claude Code | By hand only, through Career › Exchange |
| **VERA** (Virtual Executive & Reminder Assistant) | Chief of staff | Apps Script, the VERA sheet, the two dashboards | — |
| **WILLA** (Work-optional Income & Lifestyle Longevity Analyzer) | The work-optional retirement planner | Its own Google Sheet | VERA reads it; trip budgets go back and forth by paste |
| **TESS** (Tailored Employment Search Scout), "the job agent" in the code | LinkedIn job alerts, scoring, tailored materials | A Claude scheduled task, weekdays 6:36 AM ET, with its Job Agent Settings sheet; a calibration review on the 1st and 15th | Automatic, both ways, through named cells only |
| **HAILEY** (Home Alerts, Inventory & Listings Evaluator, Yours) | Zillow search against the buy box; new listings and price drops | Two Claude scheduled tasks with the HAILEY Home Search Tracker sheet | None: standalone, a roster link only |
| **ARIA** (Arrangements, Reservations & Itinerary Assistant) | Travel agent: watches and scouts fares, says when to book | A Claude scheduled task, daily at 06:52 ET, with the ARIA Travel Tracker sheet | Automatic: VERA reads ARIA's sheet; ARIA reads VERA's "ARIA Requests" tab |

**Rules that hold for every link**

- VERA never writes another assistant's data, except the named cells listed below for TESS. It never writes ARIA's sheet or WILLA's planner.
- Every read goes by header or label, never by cell position, because the other side may add or move columns. A label VERA can't find makes that piece unavailable rather than wrong.
- Sheet IDs and links live only in Config (or Script Properties), never in the repository.
- Each assistant's own instructions live in its Claude scheduled task, not in this repository. They are the other half of every contract below: a change on either side is checked against the other.
- Each assistant keeps working if VERA is down. VERA flags a link that has gone quiet instead of failing silently.

## 21.0 Everything that runs on a schedule

Times are Eastern. VERA's jobs are Apps Script triggers (`veraTriggerSpecs_` in `Code.js`; `setupTriggers` installs them and the watchdog recreates a missing one). The other assistants are Claude scheduled tasks on the same account, listed by their names there; each holds that assistant's full instructions. A Claude scheduled task can be skipped

when the account's usage limit is reached; the run is simply missed, and the health checks below notice when that matters.

| Assistant | Task | When | What it does | Results go to |
|---|---|---|---|---|
| TESS | **TESS** | Weekdays 6:36 AM | Reads LinkedIn alerts and the Companies careers pages, filters, scores, writes tailored resume, cover letter and fit note | New Tracker rows; Docs in the Applications folder; the digest email; one Run Log row per run (read by Agent health, 21.5) |
| ARIA | **ARIA daily travel check** | Daily 6:52 AM | Reads VERA's trips and requests, prices flights, judges Book now | ARIA Travel Tracker (Watchlist, Fare Log, Run Log); a Book now email |
| VERA | `morningNudge` | Daily 7:00 AM | The morning briefing, sent only when something needs you | Email |
| HAILEY | **HAILEY daily sweep** | Weekdays 7:38 AM (6:38 Central) | Searches Zillow per market, updates listings, prices and estimated monthly cost | HAILEY's tracker; an alert email only when a home crosses an alert line |
| VERA | `hourlyCheck` | Every hour | Reminders (Anticipator rules); ARIA sync from 7:30 AM, every 3 hours until 10 PM; links settled trip ideas; the watchdog; Weekend Planner on Wednesdays and the goal check-in on Mondays at `weekend_planner_hour` (8 AM) | Slack, Flags, the ARIA Requests tab |
| VERA | `checkFlightStatuses_` | Every 15 minutes | Flight status for flights within 24 hours | Slack |
| VERA | `runEmailScan_` | Every 30 minutes (when `email_parser_enabled`) | Reads booking confirmation emails | Itinerary |
| VERA | `scanUSPSMail_` | Daily 10:00 AM | Informed Delivery mail and packages | Flags, Slack |
| VERA | `scanHoaWebsite_` | Mondays 9:00 AM | The HOA website | Flags |

| Assistant | Task | When | What it does | Results go to |
|---|---|---|---|---|
| VERA | `nightlyRun`, `nightlyRunTail` | Daily 11 PM and midnight | Calendars, trips, money, the planner (WILLA), the job agent's health (TESS), patterns, new flags; the Monthly Life Review on the 1st | The VERA sheet, Flags, email |
| HAILEY | **HAILEY weekly digest** | Fridays 7:00 PM (6 PM Central) | The week's homes: available, under ceiling, new, price cuts, gone | Weekly Digest tab; the digest email |
| TESS | **TESS calibration review** | The 1st and 15th, 7:43 PM | Compares what TESS sent with what you did; read only | One email with up to 5 proposed changes; nothing changes until you approve |
| PIES | (on the work machine) | Its own schedule, not on this account | Work signal and the monthly PIES-BRIEF | The work machine only |
| WILLA | none | — | A planner sheet with no schedule of its own; VERA reads it each night | — |

## 21.1 ARIA (travel)

| From → To | What | When | How | Lands in |
|---|---|---|---|---|
| VERA → ARIA | Future trips (not driven), their budget, travellers and luggage, the itinerary, traveller preferences | ARIA's daily run | ARIA reads VERA's *Trips*, *TripMeta*, *Trip Budget*, *Itinerary* and *Traveler Items* tabs, read only | A **Draft** Watchlist row per trip; a blank ceiling is seeded from the flights budget per traveller |
| VERA → ARIA | **Watch** request for a trip's flights (home airports from Config, travellers from TripMeta, ceiling from the Trip Budget flights line) | When you tap *Ask ARIA* on a trip's Summary | VERA writes its own *ARIA Requests* tab | ARIA reads it on its morning run; it fills only blank Watchlist cells |
| VERA → ARIA | **Scout** request for a starred option of a shortlisted trip idea (at most 3 at once) | When you tap *Scout* | *ARIA Requests* | ARIA prices the idea's months and logs the cheapest date pairs under the option |

| From → To | What | When | How | Lands in |
|---|---|---|---|---|
| VERA → ARIA | Scout becomes Watch when the idea settles; new dates when a trip moves | Hourly | *ARIA Requests* | ARIA moves its row to the Trip ID and keeps the fares |
| VERA → ARIA | **Booked**: flights out and back seen in the itinerary | Each sync | *ARIA Requests* Status = Booked | ARIA stops pricing on its next run |
| VERA → ARIA | **Stay** request for a run of nights with no place to sleep: place, check-in, nights (checkout), travellers, optional ceiling per night | When you tap *Watch this stay* under *Still to book* | *ARIA Requests*, Kind = stay, Ref `STAY-<Trip ID>-<check-in>`, columns *Place* and *Ceiling Per Night* | ARIA prices hotels there on its morning run |
| VERA → ARIA | **Stay booked**: every night of the request covered by a hotel, arranged stay or cruise in the itinerary | Each sync | *ARIA Requests* Status = Booked | ARIA stops pricing that stay |
| VERA → ARIA | The household airline rule | With each request | Request columns | ARIA's search; the dashboard judges logged fares by the same rule |
| ARIA → VERA | Watchlist, Fare Log, Run Log, Settings | From 07:30, then every 3 hours until 22:00 | Read by header | Travel › ARIA, the trip Summary, Home banner, trip idea options |
| ARIA → VERA | Hotel prices: Fare Log rows whose Route starts `Hotel:` under the Trip ID | Each sync | Read by header | The stay's line under *Still to book* (best price seen) |
| ARIA → VERA | Actions: 🛫■ Book now, ■ Under your ceiling, ■ Book this stay and ■ Hotel under your ceiling (High); ■ Trip dates changed and ■■ couldn't price (Medium); watch, ceiling and airport set-up (Low) | Each sync | One flag per action, closed when it leaves the Watchlist | Flags |

| From → To | What | When | How | Lands in |
|---|---|---|---|---|
| ARIA → VERA → you | Book now and Book this stay | Once per action | Slack post plus ARIA's own email | #vera-act-now (its channel ID in Config `slack_act_now_channel_id`; #vera-notifications until it is set) |

Code: `Aria.js`. The sync is `maybeSyncAriaFlags_` on `hourlyCheck` (reads cached 10 minutes); settled ideas and moved trips are `tiLinkSettled_` and `ariaFollowTrip_`; Booked is `ariaMarkBooked_` (flights) and `ariaMarkStaysBooked_` (stays); stay requests are `webStayAriaRequest_`, and stopping a trip's flight watch never stops its stays (each stay stops by its own Ref). Flags are keyed `ARIA|<id>|<type>` by ARIA's Action Key. The Slack post is notification `aria_book_now` (on by default). The airline rule comes from Config `aria_airline_pref` and `aria_airline_premium`.

**What ARIA does on its run (from its instructions).** Requests first, then new Drafts. It prices Watching rows when their Next Check is due and Scouting rows weekly, at most 20 searches a run, soonest departure first. How often a trip is checked depends on how far away it is: more than 180 days, not yet; 90–180 days, weekly; 42–90, every 3 days; under 42, daily. It applies the airline rule, the traveller preferences (longest layover, red-eyes) and drops basic economy. **Book now** means under the ceiling and within 5% of the lowest fare it has logged for that trip (or the forecast says prices will rise), and never on a price it couldn't fully check (bags not loaded, fare type unknown). It never books, never contacts anyone but you, and never writes VERA's sheet.

**What ARIA leaves to you.** The yellow (YOU) cells in its Watchlist are yours; ARIA fills one only when it is blank and never overwrites it. ARIA never creates or stops SlickTrip price watches: *Needs watch setup* means create a SlickTrip watch for the trip and paste its Watch ID into the row (until then ARIA prices by search); *Stop SlickTrip watch* means the trip is booked, so stop the watch in SlickTrip.

**Health.** `ARIA|RUN_HEALTH` is raised when no run is logged in 26 hours or the last run failed; "VERA can't read ARIA" (High) when the tracker won't open; `ARIA|EMAIL_FAILED|<run>` when a Book now email didn't send. On a failed read only the health flag changes: nothing ARIA raised is closed. The gaps view lists future trips ARIA has no row for (trips you drive to are skipped), Watching rows with no airports or ceiling, no fare in 7 days, and requests not picked up in 2 days.

**Set-up.** Config `roster_link_aria` holds the tracker; `home_airports` gives the origins. ARIA's own preferences live in its stored instructions and change only with your approval; VERA never edits them. The manual fallbacks (Copy for ARIA, Paste ARIA's fare) are in `docs/trip-handoffs.md`. Hotels are watched only when you ask, one stay at a time.

## 21.2 WILLA (money and plans)

- **VERA reads the planner** (`planner_sheet_id`), live and by label, never writing it: Finances › Net worth and CD maturities (a flag 30 days ahead), the Summary, and planner Goals (`NetWorth.js`, `FinSummary.js`, `PlannerGoals.js`).
- **Trip budgets by paste** (`docs/trip-handoffs.md`): *Copy for WILLA* on a trip idea option sends the estimate; *Budget from WILLA* takes the figure back, with a preview.
- **The final total.** When an idea settles into a calendar trip, VERA raises a Low flag "Send the final total for … to WILLA", and keeps its reason updated with what has been booked so far (`TripActuals.js`). You send it by hand.

## 21.3 TESS (job search) and PIES (work)

Career is the only place these two meet VERA, and they never meet each other. The dashboard's Career › *How it connects* view is the live record of every crossing; `test_career_exchange` checks it lists every payload VERA reads or writes.

| From → To | What | When | How | Lands in |
|---|---|---|---|---|
| VERA → TESS | North Star constraints, current-employer exclusion, credentials held, the VERA Inbox link | On save | Automatic: VERA's own Settings cells only | Job Agent Settings › Settings |
| VERA → TESS | New Evidence | Monthly | Automatic | Wins bank doc, a new "YYYY-MM update" tab |
| TESS → VERA | Matches with track, scores, salary ranges, suggested people, links | Weekdays 6:36 AM | Automatic read of the Tracker | Career › Pipeline, Market, quarterly criteria suggestions |
| VERA → TESS | Status and Pass reason | When you set it | Automatic: the Tracker's Status and Pass reason columns, nothing else | Tracker |
| TESS → you | Daily digest with a "Triage in VERA" link | Weekdays 6:36 AM | Email | Your inbox → Career › Pipeline |
| PIES → VERA | PIES-BRIEF v1 | Monthly | Paste into Exchange (previewed, names and figures marked) | Evidence, readiness notes, Pipeline calibration |
| VERA → PIES | VERA-INTENTS v1 | Quarterly | Copy from Exchange | The PIES Goals doc, "Career Intents" |
| VERA → Claude → VERA | MATCHER-CONTEXT v1 out, ROLE-MATCH v1 back | A role found outside the alerts | Copy, then paste into Exchange | Pipeline (one entry if TESS has the same link) |

Code: `Career.js` (VERA's own Settings cells are `CAR_AGENT_OWNED_LABELS_`), `CareerCredentials.js`; the dashboard's crossings list is `CAR_CROSSINGS` in `docs/app.js`.

**What TESS does on its run (from its instructions).** It reads the LinkedIn alert emails and the careers pages on its Companies tab, filters by your settings (levels, seniority ceiling, location and pay floors, deal-breakers, CPA-required and accounting roles), verifies each posting on the employer's site, scores it out of 100 on four parts, makes a tailored resume, cover letter and fit note from the master resume, the Wins bank and *Credentials held* only, appends Tracker rows, and emails the digest. It never applies or contacts anyone, and the only thing it writes in its sheet is new Tracker rows.

**How TESS learns.** It reads the Status and *Pass reason* you set on the last 60 days of Tracker rows: Saved, Applied and Interviewing count for a kind of role, Passed (with a reason other than Other) against it, three alike before it counts. Twice a month the TESS calibration review emails proposed changes; nothing changes until you approve. VERA's Pipeline uses the same words (Saved, Applied, Interviewing, Passed and the rest; an old "Handoff" reads as Saved) and asks for one of TESS's reasons when you pass, so triaging in VERA teaches TESS directly.

**Health.** TESS: a Medium flag when the Tracker has had no new row for `career_agent_stale_weekdays` (default 2) weekdays. PIES has no automatic check: the monthly brief is a reminder, not a feed.

**Set-up.** Config `career_alert_sheet_id` (Job Agent Settings) and `career_wins_bank_doc_id`. The manual copy from the work machine is the compliance gate: no shared files and no sync between PIES and VERA. The agent never sees your package, only your floors, and internal postings are PIES's job, so your current employer is always excluded.

## 21.4 HAILEY (home search)

Nothing crosses between HAILEY and VERA except Agent health. A daily sweep (weekdays, 6:38 AM Central) searches Zillow per market, keeps its own tracker and emails only when a listing crosses an alert line; a weekly digest goes out on Fridays at 6 PM Central. VERA links to it from the roster (`roster_link_hailey`) and, for Agent health (21.5), reads its Run Log, Settings "Last run status" and Weekly Digest tabs by header. It never writes HAILEY's sheet.

## 21.5 Agent health (`AgentHealth.js`, #247)

System › Checks › **Agent health**, next to API health: whether each Claude agent ran when it should have, and what it last reported. VERA can't query Claude scheduled tasks, so each agent's health comes from what it writes to **its own** sheet, which VERA only reads.

| Agent | Where VERA looks | Flag | |
|---|---|---|---|
| ARIA | The Run Log tab in the ARIA Travel Tracker | Its own `ARIA | RUN_HEALTH` (Aria.js), not a second one |
| TESS | The Run Log tab in the Job Agent Settings sheet (Run Time (UTC), Status, Summary), written by step 11 of TESS's task. Until that tab exists, its job finds (`carAgentHealth_`), which `carCheckAgentHealth_` flags as before | `AGENT_HEALTH | TESS` |
| HAILEY | The sweep's Run Log (no Status column: errors in Notes make a run Partial; "Failed" in Settings "Last run status" fails the latest), and the Friday digest's Weekly Digest tab | `AGENT_HEALTH | HAILEY` |
| VERA | Its watchdog and heartbeats (Watchdog.js) | The watchdog's own | |

| Agent | Where VERA looks | Flag | |
|---|---|---|---|
| PIES | Not visible to VERA: always No signal, never flagged | — | |

• **Late:** no run logged for the latest scheduled run once it's due: 2 hours after the scheduled time, or the end of that day for TESS (`eod`).

• **Failed:** the latest run says Failed. **Partial** shows amber in the strip and counts as OK.

• **No signal:** nothing logged yet (not flagged), or a sheet VERA can't open (flagged).

• Each card shows the schedule, the last run, what it said, a strip of the last 7 scheduled runs (gray = expected, not logged) and a link to the sheet; the roster in System › Assistants gets a dot and a short line per agent.

• The nightly run (`checkAgentHealth_`) holds one flag per agent with `upsertKeyedFlags_`, reworded as things change and resolved on the next good run, and posts each change of level to #vera-logs. `agent_health` (GET, cached ten minutes; `refresh=1` for Check again) serves the panel.

• Schedules are Config rows, since VERA can't read Claude's: `agent_schedule_aria`, `agent_schedule_tess`, `agent_schedule_hailey`, `agent_schedule_hailey_digest`, written as `<daily|weekdays|mon,wed,…> HH:MM <time zone> [eod]`. Change them here when a task's schedule changes.

### 21.6 Config rows for the assistants

| Row | Used for |
|---|---|
| `roster_link_pies` … `roster_link_aria` | The Home roster links. WILLA and TESS fall back to `planner_sheet_id` and `career_alert_sheet_id` |
| `roster_link_aria` | Also ARIA's tracker (the old Script Property `ARIA_SHEET_ID` is read until it is moved) |
| `planner_sheet_id` | WILLA's planner |
| `career_alert_sheet_id`, `career_wins_bank_doc_id` | TESS's Settings sheet and the Wins bank doc |
| `career_agent_stale_weekdays` | Days without a TESS match before the stale flag |
| `aria_airline_pref`, `aria_airline_premium` | The airline rule sent with ARIA requests |
| `home_airports` | Origins for ARIA watch requests |
| `agent_schedule_aria`, `agent_schedule_tess`, `agent_schedule_hailey`, `agent_schedule_hailey_digest` | When each agent is scheduled, for Agent health (21.5) |

## 22. Appendix: Developer Reference

*Carried over from the old developer cheat sheet (last updated November 1, 2026). Modules added since are covered in sections 15–20.*

## Triggers setupTriggers() in Code.js

| Function | Schedule | Purpose |
|---|---|---|
| `nightlyRun` | Daily 11 PM | Main intelligence pipeline — summaries, PTO, patterns, Claude flags, Slack log |
| `morningNudge` | Daily 7 AM | Morning email digest with active flags, tasks, calendar preview |
| `hourlyCheck` | Every 1 hour | Anticipator reminder rules (ergonomic, hydration, bills, packing) + Weekend Planner on Monday |
| `checkFlightStatuses_` | Every 15 min | Flight status monitoring via AviationStack — only active when trips are within 24h of departure |
| `runEmailScan_` | Every 30 min | Travel email parser — gated: `email_parser_enabled=true`. ■ Up to 144 Claude API calls/day when enabled |

## Nightly Pipeline Steps nightlyRun() in Code.js

| Step | Function | Description |
|---|---|---|
| -1 | `escalateAgedFlags_` | Escalate unacknowledged flags aged 3d → "3d", 7d → "7d" in Escalated column |
| 0 | `writeSummarySnapshot` | Auto-populate Metrics + Summaries tabs from live data (Phase 5) |
| 0a | `syncCalendarBirthdaysToImportantDates_` | Sync upcoming birthdays from Joint Chaos calendar → Important Dates tab |
| 0a-ii | `resetChoresByCadence_` | Reset household chores checkboxes based on cadence (daily/weekly/monthly/etc.) |
| 0b | `writePTOSnapshot_` | Write PTO tab snapshot + compute VERA calendar recommendations |
| 0c | `runExplorer_` | Daily AI discovery bulletin — sends to Slack/email based on `explorer_interests` |
| 0d | `getSuppressedKeyPatterns_` | Signal Learning — load noise-suppressed patterns to filter from Claude's output |

| Step | Function | Description |
|---|---|---|
| 0e | `recordExpiredFlags_` | Signal Learning — record flags open 30+ days without action as "expired" |
| 0f | `checkPreTripBriefings_` | 48-hour pre-trip auto-briefing flags (weather, packing status, flight info) |
| 0g | `checkPostTripCapture_` | Post-trip debrief prompt — fires N days after trip end |
| 0h | (inline) | Reset Morning Routine checkboxes (Checked=false, Checked At=blank) |
| 0i | `checkGymSessions_` | Gym session check-in — scans for ended EXERCISE-tagged calendar events |
| 0j | `checkFitnessConsistency_` / `checkFitnessTravelGap_` | Weekly gym consistency check + travel gap gap detection |
| 0k | `autoRestockItems_` / `generatePantryFlags_` | Pantry: auto-restock prediction + trip-overlap flags (gated: `pantry_enabled=true`) |
| 0l | `inferCapacityMode_` | Score tomorrow's calendar load → infer busy/normal/light mode |
| 0m | `checkContracts_` | Contract expiry flags — upcoming renewals/expirations in Contracts tab |
| 0n | `checkHealthAppointments_` | Health appointment due-date checks — reads DR:-prefixed calendar events |
| 0o | `checkMonthlyReview_` | Monthly Life Review — fires on 1st of month (gated: `monthly_review_enabled=true`) |
| 0p | `resetWeekMealPlan_` | Saturday only: archive current meal week, seed next week |
| 0q | `checkCrossPatternFlags_` | Cross-domain pattern recognition — compound signals across all domains (PatternRecognition.js) |
| 1 | `getUpcomingEvents / getOpenTasks / getSummaries / getSharedInterestLedger_` | Collect live data for Claude prompt |
| 1b | `suggestDueDates` | AI-suggest due dates for undated tasks — writes back to Tasks tab |
| 2–3 | `generateFlags` | Build Claude prompt → call claude-sonnet-4-6 → parse flags JSON |
| 4 | `writeFlags` | Append flag rows to Flags tab with dedup (exact + fuzzy key matching) |

| Step | Function | Description |
|---|---|---|
| 4b | `recordFlagsGenerated_` | Signal Learning — record generated flag keys as "seen" |
| end | `sendSlackLog_` | Post nightly summary to #vera-logs (flag counts, step warnings, elapsed time) |

## Flag Sources source: values written by writeFlags()

| Source Value | Module / Step | What it generates |
|---|---|---|
| `Calendar` / `Tasks` / `Summaries` | Claude.js | Primary nightly Claude output — cross-domain analysis |
| `Health Tracker` | HealthTracker.js | Overdue / upcoming health appointments (DR: prefix events) |
| `Fitness Tracker` | Fitness.js | Weekly gym consistency and travel gap flags |
| `Gym Tracker` | GymTracker.js | Post-session check-in prompts |
| `Pattern Recognition` | PatternRecognition.js | Cross-domain compound pattern flags (all 7 patterns) |
| `Pre-Trip Briefing` | PreTripBriefing.js | 48h pre-departure briefing |
| `Post-Trip Capture` | PostTripCapture.js | Post-return memory capture prompt |
| `Email Parser` | EmailParser.js | Parsed travel confirmations from inbox |
| `Monthly Review` | MonthlyReview.js | Monthly life review flag (1st of month) |
| `Important Dates` | ImportantDates.js | Upcoming birthdays, anniversaries (30-day lead) |
| `Contracts` | Code.js (checkContracts_) | Contract renewal / expiry alerts |
| `Pantry Tracker` | Pantry.js | Purchase-history restock flags (gated) |
| `Finance` | FinancialGoals.js | Financial goal milestone / savings rate alerts |

| Urgency | Visual | Sheet Background | Meaning |
|---|---|---|---|
| High | ■ | `#fee2e2` (red tint) | Needs attention soon — @mention in Slack |
| Medium | ■ | `#fef9c3` (yellow tint) | Review this week |
| Low | ■ | `#dcfce7` (green tint) | Good to know, no urgency |

## Calendar Event Prefixes Special title prefixes parsed by VERA

| Prefix | Example Title | Detected By | Effect |
|---|---|---|---|
| `DR:` | `DR: Ahmed - Annual Physical` DR: [Person] - [Type] - [Dr. Name (opt)] | HealthTracker.js | Tracks last-visit date per person+type, computes next-due, generates flag when overdue/approaching |
| `EXERCISE` (in description) | Any title with `EXERCISE` in the event description | GymTracker.js | Logs attended gym session to Gym Log tab; generates check-in flag |
| `TRIP:` (convention) | `TRIP: Paris - CDG → ORD` | FlightStatus.js, PreTripBriefing.js | Trip tracking — pre-trip briefing, packing, itinerary, post-trip capture. Any multi-day event works. |
| `calendar_label:*` | Config row, not an event prefix | Calendar.js | Maps calendar name to a display label for Claude prompt context |

## Config Tab Keys All cfg['...'] lookups — Setting column in Config sheet

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| calendar_days_ahead | 7 | Calendar.js | Days of calendar events to fetch for nightly analysis | | | |
| task_age_threshold_days | 7 | Tasks.js / Code.js | Days before an open task is considered "neglected" | | | |
| max_flags_per_night | 8 | Claude.js | Maximum Claude-generated flags per nightly run | | | |
| morning_nudge_time | 7 | Code.js | Hour (24h) for morning nudge trigger | | | |
| snooze_default_days | 2 | WebApp.js | Default snooze duration in days | | | |
| finance_review_day | 1 | Code.js | Day of month for finance review | | | |

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| active_sources | Calendar,Tasks,Summaries | Code.js | Comma-separated data sources included in Claude prompt | | | |
| skip_calendars | Holidays in United States | Calendar.js, HealthTracker.js, GymTracker.js | Comma-separated calendar names to ignore entirely | | | |
| victoria_calendars | blank | Calendar.js, WebApp.js | Comma-separated calendar names belonging to Victoria | | | |
| calendar_label:* | none | Calendar.js | Maps calendar name to display label (e.g. `calendar_label:Ahmed = personal`) | | | |
| pto_vacation_days | 20 | PTO.js | Annual vacation allocation in days | | | |
| pto_rollover_days | 0 | PTO.js | Days carried over from prior year | | | |
| pto_personal_hours | 48 | PTO.js | Annual personal time in hours | | | |
| pto_buffer_days | 3 | PTO.js | Reserve days held back from planning calculations | | | |
| email_parser_enabled | false | EmailParser.js | Enable 30-min inbox scan for travel confirmations. ■ Up to 144 Claude calls/day | | | |

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| email_admin_enabled | false | EmailAdmin.js | Enable weekly inbox triage + follow-up tracking | | | |
| email_admin_frequency | weekly | EmailAdmin.js | `weekly` or `3days` | | | |
| pretrip_briefing_enabled | true | PreTripBriefing.js | Generate 48h pre-departure briefing flags | | | |
| pretrip_briefing_hours | 48 | PreTripBriefing.js | Hours before departure to generate briefing | | | |
| posttrip_capture_enabled | true | PostTripCapture.js | Enable post-trip memory capture prompt | | | |
| posttrip_capture_delay_days | 1 | PostTripCapture.js | Days after trip end to fire capture flag | | | |
| gym_tracker_enabled | true | GymTracker.js | Enable gym session tracking from calendar EXERCISE tags | | | |
| gym_tracker_lookback_hours | 24 | GymTracker.js | Hours to look back for ended EXERCISE calendar events | | | |
| gym_sessions_per_week | 3 | PatternRecognition.js | Target gym sessions per week for pattern detection | | | |
| fitness_enabled | false | Fitness.js | Enable weekly gym consistency check flags | | | |

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| fitness_weekly_target | 4 | Fitness.js | Target gym sessions per week for consistency flags | | | |
| fitness_low_flag_day | 4 | Fitness.js | Day of week to fire Low flag if behind (1=Sun … 7=Sat; 4=Wed) | | | |
| fitness_travel_block_time | 07:00 | Fitness.js | Start time for auto-created trip gym session blocks | | | |
| fitness_travel_block_duration | 60 | Fitness.js | Duration (minutes) for auto-created trip gym session blocks | | | |
| pacing_enabled | true | Pacing.js | Enable automatic pacing mode detection and suggestions | | | |
| pacing_flag_threshold | 3 | Pacing.js | Minimum High flags to trigger pacing mode offer | | | |
| pacing_mode_days | 7 | Pacing.js | Duration (days) for pacing mode when activated | | | |
| reminders_enabled | true | Reminders.js | Master switch for Anticipator reminder rules | | | |
| explorer_enabled | true | Reminders.js | Master switch for Explorer daily AI discovery | | | |
| explorer_interests | (built-in default) | Reminders.js | Interest topics injected into Explorer AI prompt | | | |

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| ergonomic_interval_min | 60 | Reminders.js | Ergonomic break target interval in minutes | | | |
| hydration_interval_min | 120 | Reminders.js | Hydration reminder interval in minutes | | | |
| mobility_reminder_hour | 20 | Reminders.js | 24h hour for evening mobility nudge | | | |
| weekend_planner_enabled | true | WeekendPlanner.js | Enable the Wednesday morning Weekend Planner Slack message | | | |
| weekend_planner_hour | 8 | Reminders.js, WeekendPlanner.js | Hour to fire Weekend Planner (fires from hourlyCheck on Wednesday; the Monday goal check-in uses the same hour) | | | |
| weekend_planner_lookahead_days | 21 | WeekendPlanner.js | Days ahead to look for upcoming trips/events | | | |
| monthly_review_enabled | true | MonthlyReview.js | Generate monthly life review on 1st of each month | | | |
| pantry_enabled | false | Pantry.js | Enable purchase history + auto-restock tracking | | | |
| pantry_restock_days_ahead | 7 | Pantry.js | Days ahead to predict and flag restock items | | | |

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| pantry_ema_alpha | 0.3 | Pantry.js | EMA learning rate for pantry consumption model (higher = adapts faster) | | | |
| google_tasks_enabled | true | WebApp.js | Fetch Google Tasks in dashboard and chat | | | |
| morning_routine_enabled | true | WebApp.js (status) | Show Morning Routine tab in dashboard | | | |
| takeouts_enabled | true | WebApp.js (status) | Show Takeouts tab in dashboard | | | |
| experiments_enabled | true | WebApp.js (status) | Show Experiments tab in dashboard | | | |
| wish_list_enabled | true | WebApp.js (status) | Show Wish List tab in dashboard | | | |
| wishlists_enabled | true | WebApp.js (status) | Show Christmas Wish Lists (People tab) | | | |
| dates_default_lead_time | 30 | ImportantDates.js | Default lead days for important date flags | | | |
| dates_high_urgency_days | 1 | ImportantDates.js | Days-until threshold for High urgency important date flag | | | |
| dates_medium_urgency_days | 7 | ImportantDates.js | Days-until threshold for Medium urgency important date flag | | | |

| Key | Default | Module | What it controls | | | |
|---|---|---|---|---|---|---|
| memory_log_enabled | true | Memory.js | Enable VERA memory/reflection log | | | |
| memory_snapshot_day | 0 | Memory.js | Day of week for memory snapshot (0=Sun) | | | |
| memory_log_retention_months | 12 | Memory.js | Months of memory log to retain before pruning | | | |
| monthly_disposable_income | 5000 | FinancialGoals.js | Monthly disposable income for financial goal scenario simulation | | | |
| pattern_max_flags | 2 | PatternRecognition.js | Max new pattern flags per nightly run | | | |
| pattern_dedup_days | 7 | PatternRecognition.js | Days to suppress same pattern key after it fires | | | |
| weather_location | blank | Weather.js | City name for weather ticker (e.g. `Austin, TX`) | | | |
| travel_day_briefing_enabled | true | TravelDayBriefing.js | Enable travel day briefing messages | | | |
| travel_companions | blank | TravelDayBriefing.js | Default travel companions string | | | |
| summary_sheet:* | none | Summaries.js | External sheet hook: value = `SheetID\ | TabName\ | CellRef\ | metric_name` |

## Script Properties Project Settings → Script Properties in Apps Script editor

The ids, links, channel and user ids, the email address (`morning_nudge_email`, required) and the search engine are Config rows now; see *Script Properties* above for the list and how `moveSettingsToConfig()` moves them.

| Property Key | Required? | Used By | Description |
|---|---|---|---|
| CLAUDE_API_KEY | Required | Claude.js | Anthropic API key for claude-sonnet-4-6 |
| VERA_SHEET_ID | Required | Code.js (CONFIG) | Google Sheet ID of the Life OS sheet |
| VERA_WEB_TOKEN | Required | WebApp.js | Auth token for all WebApp GET/POST requests (`?token=...`) |
| SLACK_BOT_TOKEN | Optional | Slack.js | Slack bot OAuth token (xoxb-...) |
| SLACK_ALLOWED_USER_IDS | Optional | Slack.js | Comma-separated Slack member IDs authorized to chat with VERA |
| AVIATIONSTACK_KEY | Optional | FlightStatus.js | AviationStack API key for flight status monitoring |
| WEATHER_API_KEY | Optional | Weather.js, WebApp.js | OpenWeather API key for weather ticker and destination forecasts |
| VERA_SEARCH_API_KEY | Optional | Chat.js | Serper.dev or Tavily API key for web search tool in chat |
| VERA_STATE_PACING | Runtime | Pacing.js, Away.js | Pacing and Away mode, one JSON record (written by VERA) |
| VERA_STATE_PTO_BUFFER / VERA_STATE_MAIL | Runtime | PTO.js / MailCounter.js | The buffer-day suggestion; the mail counter |
| VERA_STATE_CAPACITY_OVERRIDE / _INFERRED | Runtime | Chat.js, Slack.js / Code.js | Your capacity override; the nightly guess |
| (VERA State tab) | Runtime | FlightStatus.js, Reminders.js, WeekendPlanner.js | Flight status cache, flight-days list, discovery and weekend histories: rows of the hidden tab, not properties |
| LAST_NIGHTLY_RUN | Runtime | Slack.js (App Home) | Timestamp of last successful nightly run |
| SLACK_PINNED_MORNING_TS | Runtime | Slack.js | Slack message timestamp of pinned morning summary |

## Sheet Tabs TABS constant in Code.js — all 80+ tabs

| Constant | Tab Name | Purpose | Auto-managed? |
|---|---|---|---|
| TABS.FLAGS | Flags | All VERA-generated flags with urgency, status, escalation | Auto |
| TABS.TASKS | Tasks | Open tasks with due dates, recurring, flagged status | Manual |
| TABS.METRICS | Metrics | VERA self-monitoring counts (tasks, flags, calendar); wiped nightly | Auto |
| TABS.SUMMARIES | Summaries | External life intelligence feed — [AUTO] rows wiped/rewritten nightly | Hybrid |
| TABS.CONFIG | Config | Key/Value config rows driving all VERA behaviour | Manual |
| TABS.PROJECTS | Projects | Multi-step projects with Claude-generated subtasks | Manual |
| TABS.GOALS | Goals | Yearly goals Kanban (To Do / Doing / Done / Paused) | Manual |
| TABS.PTO | PTO | PTO planner snapshot — written nightly by writePTOSnapshot_ | Auto |
| TABS.PTO_MEMORY | PTO Memory | Stateful PTO suggestion history; declined windows blacklist | Auto |
| TABS.REMINDERS_MEMORY | Reminders Memory | Reminder cooldown log — prevents duplicate nudges | Auto |
| TABS.INTEREST_LEDGER | Shared Interests | Shared interest ledger for Ahmed + Victoria | Manual |
| TABS.BILLS | Bills | Recurring bill tracker with paid status | Manual |
| TABS.RECIPES | Recipes | Recipe list with ingredients and tags | Manual |
| TABS.MEAL_PLAN | Meal Plan | Weekly dinner planner — Saturday reset auto-archives + seeds next week | Hybrid |
| TABS.TAKEOUT_RESTAURANTS | Takeout Restaurants | Favorite takeout restaurant list | Manual |
| TABS.TAKEOUT_ITEMS | Takeout Items | Takeout menu items linked to restaurants | Manual |
| TABS.HOME_ITEMS | Home Items | Warranties + service log for home items | Manual |

| Constant | Tab Name | Purpose | Auto-managed? |
|---|---|---|---|
| TABS.IDEAS | Ideas | Braindump repo — raw thoughts + structured ideas | Manual |
| TABS.ITINERARY | Itinerary | Trip itinerary items (flights, hotels, activities) | Manual |
| TABS.TRIP_META | TripMeta | Trip context and sentiment notes per trip key | Manual |
| TABS.PACKING_ITEMS | PackingItems | Per-trip packing list items by person and category | Manual |
| TABS.COUNTRIES | Countries | Countries visited tracker | Manual |
| TABS.BUCKET_LIST | Bucket List | Travel bucket list with priority stars | Manual |
| TABS.TRIP_RECOMMENDATIONS | TripRecommendations | AI-generated trip activity and dining suggestions | Auto |
| TABS.PROCESSED_EMAILS | Processed Emails | Email parser dedup + outcome log | Auto |
| TABS.MORNING_ROUTINE | Morning Routine | Daily routine checklist — nightly checkbox reset | Hybrid |
| TABS.GYM_LOG | Gym Log | Gym session attendance log | Auto |
| TABS.PURCHASE_HISTORY | Purchase History | Pantry purchase history + consumption intelligence | Manual |
| TABS.CAREER_POSITION | Career Position | Current role snapshot | Manual |
| TABS.CAREER_GOALS | Career Goals | Long-horizon career targets | Manual |
| TABS.CAREER_PROGRESSION | Career Progression | Career timeline / roles held | Manual |
| TABS.CAREER_DEVELOPMENT | Career Development | Skills, courses, focus areas | Manual |
| TABS.CAREER_WINS | Career Wins | Professional achievement log | Manual |
| TABS.CAREER_NETWORK | Career Network | Professional relationships tracker | Manual |
| TABS.PRESCRIPTIONS | Prescriptions | Medication tracker for Ahmed + Victoria | Manual |
| TABS.CREDIT_CARDS | Credit Cards | Card metadata + ownership (11 cards) | Manual |
| TABS.CARD_REWARDS | Card Rewards | Per-card category reward rates | Manual |

| Constant | Tab Name | Purpose | Auto-managed? |
|---|---|---|---|
| TABS.CARD_PERKS | Card Perks | Monthly/annual perk tracker with last-used date | Manual |
| TABS.LOYALTY_PROGRAMS | Loyalty Programs | Points/miles balances per program | Manual |
| TABS.REWARDS_GOALS | Rewards Goals | Redemption goal tracking | Manual |
| TABS.GIFT_PEOPLE | Gift People | People list for gift ideas (Ahmed + Victoria seeded) | Manual |
| TABS.GIFT_IDEAS | Gift Ideas | Individual gift ideas linked to people | Manual |
| TABS.IMPORTANT_DATES | Important Dates | Birthdays, anniversaries, meaningful dates — synced from calendar | Hybrid |
| TABS.CHORES | Chores | Household chore checklist — cadence-based nightly reset | Hybrid |
| TABS.TRAVELER_PROFILES | Traveler Profiles | Passport + traveler profiles for visa checking | Manual |
| TABS.CONTRACTS | Contracts | Active contract + agreement tracker | Manual |
| TABS.VEHICLES | Vehicles | Vehicle tracker — oil, service, registration, tires | Manual |
| TABS.FINANCIAL_GOALS | Financial Goals | Financial goals + what-if scenario planning | Manual |
| TABS.FINANCIAL_SCENARIOS | Financial Scenarios | Saved what-if scenarios per goal | Manual |
| TABS.EMAIL_FOLLOW_UPS | Email Follow-ups | Email Admin follow-up tracking | Auto |
| TABS.BOOKS | Books | Reading list per person | Manual |
| TABS.COURSES | Courses | Courses and learning content tracker | Manual |
| TABS.SKILLS | Skills | Skill building + practice log | Manual |
| TABS.EXPERIMENTS | Experiments | Personal experiment tracker with hypothesis + outcomes | Manual |
| TABS.EXPERIMENT_CHECKINS | Experiment Checkins | Per-experiment check-in log | Manual |
| TABS.RESOURCES | Resources | Reference links + docs (Explore tab) | Manual |

| Constant | Tab Name | Purpose | Auto-managed? |
|---|---|---|---|
| TABS.BUCKET_ACTIVITIES | BucketActivities | Per-destination activity list linked to bucket list | Manual |
| TABS.WISH_LIST | Wish List | Aspirational purchase tracker per person | Manual |

## Slack Channels Slack.js — getSlackChannelId_ channel map

| Channel | Property Key | Direction | What posts there |
|---|---|---|---|
| #vera-chat | Config slack_chat_channel_id | Bidirectional | Ahmed + Victoria conversational chat. Messages processed async via CacheService queue. @mentions resolve to names via SLACK_AHMED_USER_ID / SLACK_VICTORIA_USER_ID. |
| #vera-notifications | Config slack_notifications_channel_id | Outbound only | Flag alerts with Block Kit Acknowledge + Snooze buttons. High-urgency flags @mention Ahmed via SLACK_AHMED_USER_ID. Morning nudge summary. Weekend Planner. |
| #vera-logs | Config slack_logs_channel_id | Outbound only | Nightly run summaries (flag counts, step warnings, elapsed time). WebApp errors. Flight status updates. Feature fire confirmations. |

## WebApp GET Actions doGet() in WebApp.js — all require ?token=VERA_WEB_TOKEN

| action param | Additional params | Description |
|---|---|---|
| status | — | Flag counts (total, active, high/med/low), last run date, feature flags for dashboard tab visibility, travelConfig |
| flags | filter=active (opt) | All flags, or unacknowledged+unresolved only when filter=active |
| tasks | — | Open tasks from Tasks tab |

| action param | Additional params | Description |
|---|---|---|
| `get_google_tasks` | — | Tasks from Google Tasks API (gated: google_tasks_enabled) |
| `complete_google_task` | `id` | Mark a Google Task as complete |
| `summaries` | — | Summaries tab data |
| `acknowledge` | `id` | Acknowledge a flag by ID |
| `snooze` | `id`, `days` | Snooze a flag for N days |
| `resolve` | `id` | Resolve a flag by ID |
| `complete_task` | `id` | Mark a task as Done |
| `add_task` | `task`, `dueDate`, `recurring` | Add a new task |
| `update_task` | `id`, `field`, `value` | Update a task field |
| `delete_task` | `id` | Delete a task by ID |
| `shopping` | — | Shopping list by store |
| `shopping_toggle` | store, item | Toggle shopping item purchased state |
| `shopping_add` | store, item | Add item to shopping list |
| `shopping_delete` | store, item | Delete shopping item |
| `shopping_update` | params | Update shopping item text |
| `projects` | — | All projects with subtasks |
| `complete_project_task` | `row` | Complete a project subtask |
| `add_project_task` | project, task, priority, dueDate | Add subtask to project |
| `update_project_task` | params | Update project subtask |
| `delete_project_task` | params | Delete project subtask |
| `goals` | — | Yearly goals |
| `add_goal / update_goal / delete_goal` | params | Goal CRUD |
| `interests` | — | Shared Interest Ledger |
| `pto` | — | PTO snapshot + stats |
| `budget` | — | SAT budget data |
| `bills` | — | Bills list |
| `bills_toggle` | row | Toggle bill paid status |
| `calendar_bills` | — | Calendar-derived bill events |

| action param | Additional params | Description |
|---|---|---|
| `tx_list` | — | Transaction list from Transactions sheet |
| `recent_transactions` | — | Recent transactions summary |
| `cashflow` | month, year | Monthly cashflow projection |
| `recipes` | — | Recipe list |
| `meal_plan` | week | Meal plan for a given week |
| `set_meal / update_meal_status / suggest_meals` | params | Meal plan management |
| `takeouts` | — | Takeout restaurants + items |
| `homesteward` | — | Home items + warranties + service log |
| `ideas` | — | Idea braindump + thought inbox |
| `itinerary` | tripKey | Trip itinerary items |
| `get_trip_meta / set_trip_meta` | tripKey | Trip context/sentiment notes |
| `get_packing` | tripKey | Packing list for a trip |
| `generate_packing` | tripKey, start, end | AI-generate full packing list |
| `countries` | — | Countries visited |
| `get_bucket_list` | — | Travel bucket list |
| `flight_statuses` | tripKey | Current flight statuses for a trip |
| `force_flight_statuses` | tripKey | Force-refresh flight statuses via AviationStack |
| `recommendations` | tripKey | AI trip activity/dining recommendations |
| `chat` | message, session | VERA conversational AI (Phase 4) |
| `morning_routine` | — | Morning routine checklist |
| `gym_log` | — | Gym session log |
| `career` | — | Career data (position, goals, progression, wins, network) |
| `prescriptions` | — | Medication tracker |
| `cards` | — | Credit card hub (cards, rewards, perks, loyalty, goals) |
| `get_gift_data` | — | Gift people + gift ideas |
| `get_important_dates` | — | Important dates list |

| action param | Additional params | Description |
|---|---|---|
| `get_chores` | — | Household chores with cadence + status |
| `financial_goals` | — | Financial goals + scenarios |
| `get_vehicles` | — | Vehicle tracker data |
| `get_contracts` | — | Active contracts |
| `health_appointments` | — | Health appointments derived from DR: calendar events |
| `get_growth` | — | Books, courses, skills (Growth / Explore tab) |
| `get_experiments` | — | Personal experiments + check-ins |
| `get_resources` | — | Reference links and docs |
| `get_wish_list` | — | Aspirational wish list |
| `get_pacing_status` | — | Current pacing + Away mode status |
| `away_status` | — | Away mode, and what it has held |
| `agent_health` | `refresh=1` (optional) | Agent health: each agent's level, last run, strip and source |
| `set_notif_while_away` | `key`, `value` | A notification's While away setting (send / hold / due) |
| `get_visa_requirements` | origin, destination | Visa requirements for a traveler profile |
| `dest_weather` | location | Weather forecast for destination |
| `tx_aliases / set_tx_alias` | params | Transaction merchant alias management |

## WebApp POST Actions doPost() in WebApp.js — JSON body required

| action | Body params | What it does |
|---|---|---|
| `chat` | `message`, `session` | Process a chat message (same as GET chat) |
| `acknowledge` | `id` | Acknowledge a flag by ID |
| `snooze` | `id`, `days` | Snooze a flag for N days |
| `resolve` | `id` | Resolve a flag by ID |
| `add_takeout_restaurant` | name, cuisine, phone, website, rating, notes | Add a takeout restaurant |

| action | Body params | What it does |
|---|---|---|
| delete_takeout_restaurant | name | Delete a takeout restaurant |
| add_takeout_item | restaurant, item, description, rating, notes | Add a menu item to a restaurant |
| delete_takeout_item | restaurant, item | Delete a takeout menu item |
| log_purchase_run | items array | Log a grocery/shopping run to Purchase History |
| Note: Slack Events API POSTs (type: event_callback) and Slack form payloads (Block Kit interactions) are handled automatically without auth token — detected by body.type or e.parameter.payload. | | |

## Cross-Domain Patterns PatternRecognition.js — all 7 patterns

| Pattern Key | Trigger Conditions | Urgency |
|---|---|---|
| cross_high_stress_compound | overdueTasks ≥ 3 AND activeHighFlags ≥ 2 AND gymSessionsThisWeek = 0 | High |
| cross_goal_behaviour_drift | activeGoals ≥ 2 AND gymSessionsThisWeek < (gymTarget × 0.5) AND overdueTaskCount ≥ 2 | Medium |
| cross_social_calendar_gap | calendarEventsThisWeek ≤ 1 AND pacingMode = false AND vacationMode = false | Low |
| cross_overload_no_pacing | intensityLevel = 'high' AND pacingMode = false AND activeHighFlags ≥ 2 AND overdueTaskCount ≥ 3 | Medium |
| cross_meal_chaos | takeoutRatio ≥ 0.70 (≥70% of meals are takeout/eating out) AND overdueTaskCount ≥ 2 | Low |
| cross_backlog_accumulation | neglectedTaskCount (age ≥ 14 days) ≥ 5 AND calendarEventsThisWeek ≥ 3 | Medium |
| cross_health_neglect | overdueHealthAppts ≥ 2 AND gymSessionsThisWeek = 0 | Medium |

## Health Appointment Default Intervals HEALTH_APPT_INTERVAL_DEFAULTS_ in HealthTracker.js

| Type (exact match keys) | Interval (months) |
|---|---|
| `physical` / `annual physical` | 12 |
| `dental` / `dentist` / `dental cleaning` / `teeth cleaning` | 6 |
| `eye exam` / `optometrist` / `ophthalmologist` | 12 |
| `dermatology` / `dermatologist` | 12 |
| `gynecology` / `obgyn` / `gynaecology` | 12 |
| `therapy` / `therapist` | 1 |
| `cardiology` / `cardiologist` | 12 |
| `endocrinology` / `endocrinologist` | 12 |
| `allergist` / `allergy` | 12 |
| `chiropractor` / `chiropractic` | 1 |
| `urgent care` | 0 — never flagged (episodic) |
| Any other type | 12 (fallback) |